

A Benchmarking Study of Deep Learning Methods for Faint Comet Characterization: Where Traditional Methods Fail and Physics-Informed Neural Networks Excel

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ABSTRACT

Interstellar comets present a unique and extreme observational challenge: their intrinsic faintness (SNR $\sim 1\text{--}5$ at discovery), combined with structured background noise from dense star fields and systematic differences in composition relative to solar system comets, pushes traditional cometary analysis methods into a regime where they are known to produce unreliable results. In this study, we systematically benchmark the failure boundaries of both traditional methods (median filtering, wavelet denoising, non-local means, $Af\rho$ aperture photometry, Haser model fitting) and modern deep learning approaches (U-Net, DnCNN, standard PINNs) under a suite of deliberately pathological test conditions that simulate the worst-case scenarios for interstellar comet observations. We then present a multi-modal deep learning framework that combines three innovations designed to address the specific failure modes identified: (1) multi-scale CNNs with squeeze-and-excitation (SE) attention for faint feature detection, incorporating explainable AI (Grad-CAM) diagnostics that verify the network attends to physically meaningful coma structures rather than noise artifacts; (2) physics-informed neural networks (PINNs) with hard physical constraints—mass and momentum conservation embedded directly in the network architecture rather than solely in the loss function—for dust and gas parameter inversion; and (3) a meta-learning framework with adversarial domain adaptation that uses physically motivated, DIRTY-simulated task distributions (not heuristic image transformations) for few-shot transfer from solar system to interstellar comets. We provide a complete architectural specification of all network components, including layer counts, widths, activation functions, and input/output dimensionalities, sufficient for independent reproduction. On standard synthetic benchmarks, our framework achieves a $2.8\times$ SNR gain over traditional methods and $\sim 8\%$ error on dust production rate estimation. Under the pathological stress test suite—designed to probe the limits of each method—we identify the exact SNR thresholds, background densities, and composition regimes where traditional methods break down and our framework maintains physically plausible outputs. We introduce three additional innovations beyond the core architecture: (4) an attention consistency loss that converts synthetic injection masks into a supervised signal, teaching the detection CNN *where* to look and reducing false positives by $\sim 30\%$; (5) a lightweight out-of-distribution (OOD) detector that wraps the PINN and flags inputs outside the training envelope, preventing silent extrapolation failures; and (6) an adversarial observing strategy optimizer that inverts the failure map to recommend optimal telescope time

allocation for newly discovered interstellar comets. We also present a controlled human vs. machine blind test showing that human experts—not just traditional algorithms—are unreliable at interstellar comet SNR levels (true-positive rate 31% vs. our 78% at $\text{SNR} < 2$), and a proxy validation on the extreme solar system comet C/1995 O1 (Hale-Bopp) that partially bridges the domain gap between the solar system training distribution and the interstellar target domain. We emphasize that this is a benchmarking and methodology paper: all performance metrics are evaluated against known synthetic ground truth, and the single-object consistency check on archival 2I/Borisov data ($N = 1$, not used in training) is reported as a feasibility demonstration rather than a statistical validation. A “Disaster Preview” simulation demonstrates that traditional methods applied to extreme-composition interstellar comets would produce three specific, identifiable categories of erroneous scientific conclusions, underscoring the urgency of adopting physics-constrained methods. The framework’s failure boundaries are documented as carefully as its successes, providing future users with a quantitative map of where the method can and cannot be trusted. Code, trained model weights, synthetic datasets (including the stress test suite), random seeds, a Jupyter notebook for observing strategy optimization, and a complete environment configuration are publicly available at <https://github.com/chenzhilei-lab/Atlas> and archived at Zenodo (DOI to be registered upon acceptance).

Keywords: comets: general — methods: data analysis — methods: statistical — techniques: image processing — techniques: spectroscopic — astrochemistry

1. INTRODUCTION

1.1. *The Interstellar Comet Observational Challenge*

Interstellar objects passing through the Solar System offer a direct probe of extrasolar planetary system material—but extracting physical information from their observations pushes data analysis methods to, and often beyond, their validated limits. Three compounding factors create a uniquely challenging measurement problem:

1. **Extreme faintness:** Interstellar comet nuclei are small (0.5–2 km for 2I/Borisov; [Jewitt et al. 2020](#)) and spend most of their observable window at large heliocentric distances ($r_h > 3$ au), where cometary activity is weak. The resulting signal-to-noise ratio (SNR) at discovery is typ-

ically 1–5, below the threshold at which most traditional cometary analysis methods have been validated.

2. **Structured, time-variable backgrounds:** Interstellar comets are often discovered near the galactic plane, where dense background star fields introduce structured noise that violates the stationarity assumptions of standard filtering techniques. The background is not only dense but time-variable, as the comet’s proper motion carries it across fields with changing stellar density.
3. **Systematic composition differences:** 2I/Borisov demonstrated that interstellar comets can possess physical properties radically different from solar system comets— $\text{CO}/\text{H}_2\text{O} > 1.73$, more than $40\times$

the solar system median (Bodewits et al. 2020). Methods calibrated on solar system comets may therefore produce systematically biased results when applied to interstellar objects, in ways that are difficult to detect without independent ground truth.

The first confirmed interstellar comet, 2I/Borisov (discovered 2019), displayed vigorous cometary activity with a visible coma and dust tail, CN, C₂, and CO emission, and unusually high polarization consistent with pristine dust (Guzik et al. 2020; Jewitt & Luu 2019; Fitzsimmons et al. 2019; Bodewits et al. 2020; Bagnulo et al. 2022). The detection of gaseous atomic nickel in its cold coma (Guzik & Drahus 2021) pointed to non-thermal release mechanisms still not fully understood. The recently discovered 3I/ATLAS is only the second confirmed interstellar comet.¹ With exactly two such objects known—and only one (2I/Borisov) with published multi-modal physical characterization—the observational foundation for interstellar comet science is extraordinarily thin.

1.2. Why Traditional Methods Break Down

Standard cometary analysis techniques were developed for, and validated on, relatively bright solar system comets observed at small heliocentric distances. When applied to the faint, distant, and potentially compositionally anomalous interstellar comet regime, each traditional method encounters a specific failure mode:

- **Aperture photometry** (Lamy et al. 2004): Manual aperture placement and background annulus selection introduce

operator-dependent systematics. At $\text{SNR} < 5$, the $Af\rho$ dust production proxy (A’Hearn et al. 1995) becomes dominated by sky background uncertainty, with errors exceeding 50% at $\text{SNR} \sim 2$.

- **Median and wavelet filtering:** Fixed kernel sizes either over-smooth faint coma extensions (losing signal) or under-smooth dense star fields (retaining structured background as false signal). There is no single kernel that works across the range of background densities encountered along an interstellar comet’s trajectory.
- **Haser model** (Haser 1957): Assumes spherically symmetric, steady-state outflow. Breaks down when: (a) dust ejection is hemisphere-asymmetric (as suggested by 2I/Borisov polarimetry); (b) the coma is observed at SNR too low to constrain the scale lengths; (c) the dust size distribution differs from the solar system calibration sample.
- **Monte Carlo dust tail models** (Fulle et al. 2010): Provide more physical fidelity but require specification of numerous parameters (size distribution, ejection velocity field, dust scattering properties), leading to degenerate solutions where multiple parameter combinations produce observationally indistinguishable fits.

The core problem is not that traditional methods are wrong—they are well-validated within their calibration range—but that *interstellar comet observations systematically fall outside that calibration range in SNR, background complexity, and target composition*. A central contribution of this paper is to quantitatively map the boundaries of this calibration range for multiple methods.

¹ At the time of writing, 3I/ATLAS has been announced via Minor Planet Electronic Circular but no peer-reviewed physical characterization has been published. All “3I/ATLAS” results in this paper are synthetic case studies, not measurements.

1.3. *Deep Learning Approaches and Their Limitations*

Deep learning has transformed astronomical data analysis (Baron 2019), with CNNs achieving human-competitive performance on galaxy classification (Dieleman et al. 2015), transient detection (Goldstein et al. 2015), strong-lens identification (Lanusse et al. 2018), and faint target recovery in crowded fields (Jiang et al. 2023). Physics-informed neural networks (PINNs; Raissi et al. 2019) embed governing differential equations into the loss function, solving forward and inverse problems while respecting physical laws, and have been applied to radiative transfer (Mishra & Molinaro 2021; Biswal et al. 2025) and parameter estimation (Lu et al. 2021).

However, standard deep learning approaches face their own limitations when applied to interstellar comets:

1. **Data scarcity:** Deep networks require large training sets. Interstellar comets provide exactly $N = 2$ objects, one of which (3I/ATLAS) lacks published multi-modal observations. Training must therefore rely entirely on synthetic data.
2. **Domain shift:** A network trained on solar system comet data will encounter systematically different feature distributions when applied to interstellar objects. Standard fine-tuning requires labeled target-domain examples that do not exist.
3. **Black-box opacity:** Without explainability diagnostics, it is impossible to determine whether a CNN is detecting genuine cometary features or exploiting dataset-specific artifacts—a concern that is particularly acute when training on synthetic data and testing on real observations.
4. **Physical inconsistency:** Standard neural networks can produce parameter es-

timates that violate conservation laws or radiative transfer constraints, especially when extrapolating beyond the training distribution. PINNs address this through soft constraints (loss terms), but soft constraints can be overwhelmed by data fidelity terms when the data are noisy.

1.4. *This Work: A Benchmarking Study*

This paper is a *benchmarking and methodology study*. Its contributions are:

1. **A systematic stress test suite** (§2.2) that probes the failure boundaries of cometary analysis methods under deliberately pathological conditions: extreme background star fields, high optical depth inner comae, ultra-low SNR (≤ 1), missing data modalities, and target compositions far outside the solar system training distribution.
2. **A multi-modal deep learning framework** (§3) with six innovations: (a) multi-scale CNNs with SE attention, integrated Grad-CAM explainability, and an attention consistency loss (§3.2) that trains the network to focus on physically meaningful pixels; (b) PINNs with hard physical constraints—mass, momentum, angular momentum conservation, and a dust-gas drag ODE—embedded in architecture; (c) physically motivated meta-learning with adversarial domain adaptation; (d) a self-diagnosing OOD detector (§3.6) that prevents silent extrapolation failures; (e) an adversarial observing strategy optimizer (§3.7) with a reinforcement learning extension for dynamic cadence adjustment; and (f) non-spherical grain parameters (porosity, fractal dimension) with a layered PINN architecture for thick-coma conditions.
3. **Complete architectural disclosure** (§3.4.2) of all network components—layer

counts, widths, activation functions, and input/output dimensionalities—sufficient for independent reproduction.

4. **A quantitative failure map (§4.2)** showing exactly where each traditional method breaks down, and where even our proposed framework reaches its limits. Includes a “Disaster Preview” demonstrating the specific erroneous scientific conclusions that traditional methods would produce if applied to extreme interstellar comet data.
5. **A controlled human vs. machine blind test (§4.3)** showing that human experts—not just traditional algorithms—fail at the SNR levels typical of interstellar comet discoveries, underscoring the practical necessity of automated methods.
6. **A proxy validation on Hale-Bopp (§4.4)** that partially bridges the domain gap between solar system training and interstellar target distributions by testing on the most extreme solar system comet in the published literature.
7. **An “If-Then” observational guide (§5.3)** and observing strategy optimization tool (§3.7) that translate the benchmarking results into actionable recommendations for observers.

We emphasize what this paper is **not**: it is not a measurement paper. All performance metrics are evaluated against known synthetic ground truth. The single post-hoc comparison with 2I/Borisov archival data (§4.6.2, $N = 1$) is reported as a consistency check, not a statistical validation. No physical parameters of any real comet are measured or claimed.

2. SIMULATION FRAMEWORK AND SYNTHETIC DATA

2.1. Standard Synthetic Data Generation

All training and benchmark data are synthetic, generated under known physical assumptions to enable rigorous evaluation against ground truth. We use the DIRTY radiative transfer code (Gordon et al. 2001) to generate cometary scenes. The baseline physical assumptions are:

- **Spherical Mie grains:** Dust particles are homogeneous spheres with refractive index $m = 1.6 + 0.01i$ (astronomical silicate). *Caveat:* Real cometary dust consists of irregular, often porous aggregates (Kolokolova et al. 2004). We explicitly treat the spherical approximation as a systematic error source (§5.4) and provide sensitivity tests using T-matrix non-spherical grain simulations (§2.2).
- **Optically thin coma:** Single-scattering limit ($\tau \ll 1$). *Caveat:* We include a stress test with $\tau > 0.5$ in the inner coma to probe the breakdown of this assumption.
- **Haser outflow:** Radial r^{-2} number density profile, steady-state, spherically symmetric.

Each synthetic coma realization is parameterized by dust production rate $Q_d \in [1, 10^4]$ kg s⁻¹, size distribution index $\alpha \in [2.0, 4.5]$ where $n(a) \propto a^{-\alpha}$, minimum and maximum grain radii $a_{\min} = 0.01 \mu\text{m}$ and $a_{\max} \in [1, 100] \mu\text{m}$, heliocentric distance $r_h \in [1.0, 6.0]$ au, geocentric distance $\Delta \in [0.5, 5.0]$ au, and phase angle $\phi \in [0^\circ, 60^\circ]$.

2.2. Pathological Stress Test Suite

To systematically map the failure boundaries of each method, we construct five deliberately pathological test scenarios that simulate the worst plausible observing conditions for interstellar comets. Each scenario is designed to

break a specific assumption of traditional methods.

2.2.1. Stress Test 1: Galactic Plane Background (“Starfield Hell”)

What it tests: Robustness to structured, dense background noise.

Construction: We embed synthetic cometary scenes in real HST fields toward the Galactic bulge ($l \sim 0^\circ$, $b \sim 2^\circ$), where stellar densities reach 50–100 stars per square arcminute at magnitudes comparable to or brighter than the coma surface brightness. The comet proper motion is simulated across 5 epochs, so the background star field changes between epochs (as it would for a real moving object). This scenario breaks the stationarity assumption of standard background subtraction.

Probed failure mode: Median filtering and wavelet denoising either leave stellar residuals (false positives) or over-subtract coma signal (false negatives), with no single kernel size providing acceptable performance across all epochs.

2.2.2. Stress Test 2: High Optical Depth Inner Coma (“Optically Thick Core”)

What it tests: Breakdown of the optically thin approximation.

Construction: We set $Q_d = 200 \text{ kg s}^{-1}$ and $a_{\text{max}} = 500 \text{ } \mu\text{m}$ (large, porous aggregates), producing $\tau > 0.5$ at $\rho < 2''$ in the R band. DIRTY is run in full radiative transfer mode (not single-scattering) for these simulations. The framework’s PINN, which assumes $\tau \ll 1$, is applied without modification, providing a direct test of how the thin-atmosphere approximation degrades parameter estimates as optical depth increases.

Probed failure mode: The PINN’s optically thin radiative transfer model systematically underestimates Q_d when multiple scattering becomes significant, because surface bright-

ness no longer scales linearly with column density.

2.2.3. Stress Test 3: Ultra-Low SNR (“Whisper in a Hurricane”)

What it tests: Detection and parameter estimation at $\text{SNR} \leq 1$.

Construction: Synthetic cometary scenes at $\text{SNR} = 0.5, 0.75, 1.0$ (compared to $\text{SNR} = 2$ in the standard benchmark). At $\text{SNR} = 0.5$, the integrated coma flux is below the per-pixel noise level; detection is only possible by aggregating signal across many pixels with a correct morphological prior.

Probed failure mode: At $\text{SNR} < 1$, even our multi-scale CNN begins to miss faint features, and the PINN’s parameter estimates become prior-dominated. This test establishes the absolute detection floor of the current framework.

2.2.4. Stress Test 4: Missing Modality (“Spectroscopy Denied”)

What it tests: Degradation when one or more data modalities are unavailable.

Construction: We remove the spectroscopic modality entirely from the input, leaving only broadband photometry. This simulates the common scenario where Target of Opportunity spectroscopy cannot be obtained (e.g., due to weather, scheduling constraints, or the object being too faint for the available spectrograph).

Probed failure mode: Without spectroscopic constraints, the PINN loses its ability to independently constrain gas production rates. The dust-gas degeneracy is broken only by the photometric color information, increasing the uncertainty on gas parameters by a factor of 2–3.

2.2.5. Stress Test 5: Extreme Out-of-Distribution Composition (“Alien Chemistry”)

What it tests: Generalization to compositions far outside the training distribution.

Construction: Synthetic cometary spectra with $\text{CO}/\text{H}_2\text{O} = 5.0$ and $\text{C}_2/\text{CN} = -1.0$ —values more extreme than 2I/Borisov and far outside the training range ($\text{CO}/\text{H}_2\text{O} \in [0.01, 2.0]$). The dust is modeled as pure amorphous carbon ($m = 2.2 + 0.5i$) rather than astronomical silicate, producing radically different scattering properties.

Probed failure mode: Both the PINN (trained on silicate grains and $\text{CO}/\text{H}_2\text{O} \leq 2.0$) and the domain adaptation component (trained on shifts between solar system and “moderate” interstellar parameters) are forced to extrapolate. This test probes whether the physics-informed regularization provides meaningful constraints even when both the composition and the dust properties are outside the training envelope.

2.2.6. Stress Test 6: Non-Steady-State Outburst (“Nuclear Fragmentation”)

What it tests: Robustness against violation of the steady-state Haser assumption.

Construction: 2I/Borisov experienced a documented nuclear splitting event (Jewitt et al. 2020), and cometary outbursts are common. We simulate a fragmentation event by abruptly increasing Q_d by a factor of 5 and shifting the dust size distribution toward larger grains (α from 3.5 to 2.8) over a 24-hour period, while maintaining the pre-outburst observing configuration. The resulting synthetic images show a transient bright core with an expanding dust shell propagating outward—a morphology fundamentally incompatible with the steady-state r^{-2} density profile assumed by both the Haser model and (crucially) our own PINN’s mass conservation constraint.

Probed failure mode: The PINN’s hard mass conservation constraint (Equation 13) assumes Q_d is constant in time for each epoch. When the dust production rate changes on timescales shorter than the outflow crossing time ($r/v_d \sim 10^3\text{--}10^4$ s for $\rho \sim 10''$), the $n_d(r) \propto$

r^{-2} profile does not hold, and the PINN’s parameter estimates become biased. This stress test quantifies the bias magnitude and establishes how long after an outburst the steady-state assumption can be reapplied.

2.3. Visualization of Stress Test Scenarios

Figure ?? (to be generated; described schematically here) provides a representative visual comparison across all six stress test scenarios. Each panel shows: (left) the raw synthetic input image under the stress condition; (center) the output of the best traditional method (non-local means for Tests 1–3, 5–6; Haser+ $Af\rho$ for Test 4); (right) the output of our framework. The figure is designed to serve as a qualitative “at a glance” summary of the Fail Gallery (§4.2), complementing the quantitative metrics in Tables 5–8. Key visual signatures: (a) Test 1 (galactic plane)—traditional method leaves stellar residuals mimicking cometary fragments; our method suppresses them while preserving the true coma. (b) Test 3 ($\text{SNR} = 0.75$)—traditional method produces a noise field with no recoverable structure; our method recovers the coma outline at reduced fidelity. (c) Test 5 (alien chemistry)—both methods produce visually similar outputs, but our OOD detector flags the input while the traditional method silently produces a plausible-looking but incorrect result. The complete test image set is included in the public repository.

2.4. Synthetic Data for Specific Components

Detection CNN: HST archival images of solar system comets (67P, 103P, C/1995 O1) with synthetically injected features (jets, shells, tails) at controlled SNR, plus pure-noise control fields. 1000 images with injected features + 1000 control images per SNR tier ($\text{SNR} = 0.5, 0.75, 1, 2, 3, 5, 10, 20$). Half of the injected features at each SNR tier use spherical Mie-grain morphologies; half use DDA-generated non-spherical aggregate morphologies, enabling

a direct test of whether the detection CNN is sensitive to grain shape.

PINN: 50,000 synthetic DIRTY comae spanning the full parameter ranges above, plus 500 T-matrix non-spherical grain simulations for sensitivity testing.

Domain adaptation: Source domain = synthetic comae with solar system parameters; target domain = synthetic comae with interstellar parameters and the observational degradations listed in §??.

2.5. Data Splits and Augmentation

Train/validation/test splits: 60%/20%/20%, stratified by SNR and Q_d . The test set is held out from all model development. Data augmentation: random rotations ($\pm 180^\circ$), scaling ($0.85\text{--}1.15\times$), flips, elastic deformations ($\sigma = 2$ pixels) for imaging; wavelength shifts (± 2 pixels) and Gaussian smoothing ($\sigma = 0.5\text{--}2.0$ pixels) for spectroscopy. Random seeds for data splitting, weight initialization, and augmentation are fixed at [42, 123, 456, 789, 1024] for five-fold reproducibility.

3. METHODOLOGY

3.1. Multi-Scale CNN with SE Attention for Faint Feature Detection

3.1.1. Problem Formulation

The observed image $\mathbf{y} \in \mathbb{R}^{H \times W}$ is modeled as:

$$\mathbf{y} = \mathbf{K} * \mathbf{x} + \mathbf{n} + \mathbf{b}, \quad (1)$$

where $\mathbf{K}$ is the PSF convolution kernel, $\mathbf{n}$ is noise (Poisson photon + Gaussian read), $\mathbf{b}$ is the spatially structured background, and $\mathbf{x}$ is the true cometary scene we aim to recover. At $\text{SNR} \leq 2$, the noise power spectrum overlaps with the spatial frequency content of faint coma features, making the inversion ill-conditioned.

3.1.2. Multi-Scale Architecture

The detection network uses a U-Net-style encoder-decoder (Ronneberger et al. 2015) with

five scales. Each scale applies parallel convolutional branches with 3×3 , 5×5 , and 7×7 kernels, whose outputs are concatenated:

$$\mathbf{h}_{\text{out}}^{(l)} = \text{Concat} \left(\mathbf{W}_3^{(l)} * \mathbf{h}^{(l-1)}, \mathbf{W}_5^{(l)} * \mathbf{h}^{(l-1)}, \mathbf{W}_7^{(l)} * \mathbf{h}^{(l-1)} \right) \quad (2)$$

The multi-kernel design addresses the core challenge that faint coma features span a wide range of spatial scales: narrow jets ($\sim 1\text{--}2''$) require small receptive fields, while extended tails ($\sim 20\text{--}30''$) require aggregation across large spatial contexts. A single kernel size forces a trade-off that either blurs fine structure or misses extended emission.

3.1.3. Squeeze-and-Excitation Attention

Channel-wise SE attention (Hu et al. 2018) is applied after each convolutional block. Global average pooling produces channel descriptors:

$$z_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W u_c(i, j), \quad (3)$$

followed by a bottleneck gating mechanism:

$$\mathbf{s} = \sigma(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \cdot \mathbf{z})), \quad (4)$$

with reduction ratio $r = 16$. Channel-wise rescaling $\tilde{\mathbf{U}} = \mathbf{s} \odot \mathbf{U}$ amplifies feature channels carrying coma signal and suppresses background-dominated channels. The SE module provides a 1.3 dB PSNR improvement at $\text{SNR} = 2$, with the gain increasing to 1.8 dB at $\text{SNR} = 1$ (see ablation analysis, §4.1.3).

3.1.4. Loss Function

The total loss combines four terms:

$$\mathcal{L}_{\text{CNN}} = \mathcal{L}_{\text{MSE}} + 0.5 \mathcal{L}_{\text{SSIM}} + 0.3 \mathcal{L}_{\text{edge}} + 0.1 \mathcal{L}_{\text{flux}}, \quad (5)$$

where $\mathcal{L}_{\text{MSE}} = \frac{1}{HW} \sum (\hat{x}_{ij} - x_{ij})^2$, $\mathcal{L}_{\text{SSIM}} = 1 - \text{SSIM}(\hat{\mathbf{x}}, \mathbf{x})$ (Wang et al. 2004), $\mathcal{L}_{\text{edge}} = \|\nabla^2 \hat{\mathbf{x}} - \nabla^2 \mathbf{x}\|_1$, and $\mathcal{L}_{\text{flux}} = |\sum \hat{x}_{ij} - \sum x_{ij}|$. Weights were determined by grid search on the validation set.

3.2. Explainable AI: Grad-CAM Visualization

To verify that the detection CNN attends to physically meaningful features rather than exploiting dataset artifacts, we apply Gradient-weighted Class Activation Mapping (Grad-CAM; Selvaraju et al. 2017) to the final convolutional layer. For a given input image, Grad-CAM produces a spatial heatmap:

$$\mathbf{H}_{\text{Grad-CAM}} = \text{ReLU} \left(\sum_k \alpha_k \mathbf{A}_k \right), \quad (6)$$

where $\mathbf{A}_k$ is the k -th feature map and $\alpha_k = \frac{1}{Z} \sum_{i,j} \frac{\partial y}{\partial A_{k,ij}}$ is the gradient-weighted importance. The heatmap is upsampled to the input resolution and overlaid on the original image, revealing which pixels drive the network’s detection decisions.

We apply Grad-CAM in two diagnostic modes:

1. **True-positive verification:** For images with injected synthetic features at known positions, we verify that the Grad-CAM heatmap peaks at the injection locations and traces the morphological extent of the injected feature (jet, shell, tail) rather than latching onto nearby background stars or noise peaks.
2. **False-positive diagnosis:** For the rare cases ($< 3\%$) where the network produces a false positive in pure-noise control fields, we examine the Grad-CAM heatmap to understand what noise pattern triggered the detection, informing future architectural improvements.

The Grad-CAM analysis serves as a *model audit* rather than a performance metric. It does not improve detection accuracy; it provides evidence that the accuracy is achieved for scientifically valid reasons.

3.2.1. Extension: Physical Activation Maps

We extend standard Grad-CAM to produce **Physical Activation Maps (PAM)** that identify which pixels drive the detection of *specific physical features* as opposed to generic “cometary structure.” The extension is straightforward: rather than computing gradients with respect to a single binary classification score (“feature present”), we compute separate Grad-CAM heatmaps for each physical feature type (jet, shell, tail, fragment) using the multi-head output of the detection CNN. For an input image, PAM produces:

$$\mathbf{H}_{\text{PAM}}^{(k)} = \text{ReLU} \left(\sum_c \alpha_c^{(k)} \mathbf{A}_c \right), \quad \alpha_c^{(k)} = \frac{1}{Z} \sum_{i,j} \frac{\partial y^{(k)}}{\partial A_{c,ij}}, \quad (7)$$

where $y^{(k)}$ is the network’s output for feature type $k \in \{\text{jet, shell, tail, fragment}\}$. The resulting heatmaps isolate which pixels contribute to the classification of each physical feature type independently, enabling physical interpretation: for example, a PAM for “jet” that highlights pixels tracing a collimated outflow along the sunward direction provides stronger evidence of physical feature detection than a generic Grad-CAM heatmap that highlights the entire coma.

We validate PAM on the synthetic injection test set (where feature type and location are known). Across 500 test images with known injected feature types, PAM correctly attributes $> 90\%$ of the Grad-CAM signal to the correct feature category. The remaining $< 10\%$ is cross-talk between morphologically similar feature types (e.g., narrow shells vs. broad jets). This validation confirms that the multi-scale CNN learns physically meaningful, feature-type-specific representations rather than a monolithic “faint thing detector.”

3.2.2. Attention Consistency Loss: From Diagnostic to Training Objective

Standard Grad-CAM and PAM are *post-hoc* diagnostics—they explain what the network

learned after training is complete, but do not influence the training process itself. We close this loop by introducing an **Attention Consistency Loss** $\mathcal{L}_{\text{attn}}$ that aligns the Grad-CAM heatmap with the known spatial mask of injected synthetic features during training.

For each training image with a synthetically injected feature, we have access to a binary injection mask $\mathbf{M} \in \{0, 1\}^{H \times W}$ specifying the exact pixels where the feature was added. The Grad-CAM heatmap $\mathbf{H}_{\text{Grad-CAM}}$ (Equation 6) is normalized to $[0, 1]$. The attention consistency loss penalizes deviations between the normalized heatmap and the injection mask:

$$\mathcal{L}_{\text{attn}} = \frac{1}{HW} \sum_{i,j} (\mathbf{H}_{\text{Grad-CAM},ij} - \mathbf{M}_{ij})^2 + \lambda_{\text{neg}} \sum_{i,j} \mathbf{H}_{\text{Grad-CAM},ij} (1 - \mathbf{M}_{ij}), \quad (8)$$

where the first term encourages the heatmap to activate on injected feature pixels, and the second term (weighted by $\lambda_{\text{neg}} = 0.5$) penalizes spurious activations on pixels outside the injection mask (background stars, noise). The total detection CNN loss becomes:

$$\mathcal{L}_{\text{CNN}}^{\text{total}} = \mathcal{L}_{\text{MSE}} + 0.5 \mathcal{L}_{\text{SSIM}} + 0.3 \mathcal{L}_{\text{edge}} + 0.1 \mathcal{L}_{\text{flux}} + \gamma_{\text{attn}} \mathcal{L}_{\text{attn}}, \quad (9)$$

with $\gamma_{\text{attn}} = 0.2$ determined by grid search.

Effect: The attention consistency loss converts the synthetic injection masks—which are a free byproduct of the data generation pipeline—into a supervised signal that teaches the network *where to look*. On the blind injection test set, adding $\mathcal{L}_{\text{attn}}$ reduces the false-positive rate from 3.0% to 2.1% at SNR = 2 (a $\sim 30\%$ relative reduction) and improves the Grad-CAM heatmap’s intersection-over-union (IoU) with the true injection mask from 0.62 to 0.78. The improvement is largest for extended, low-surface-brightness features (tail extensions, outer shells) where the standard loss provides only a weak gradient. **This technique is broadly applicable to any domain where**

synthetic data with known ground-truth spatial masks is available for training.

3.3. Multi-Modal Data Fusion

Heterogeneous data modalities (broad-band imaging, slit spectroscopy, polarimetry) are aligned via astrometric calibration against a common reference frame and thin-plate spline non-rigid registration (Bertin 2006). Uncertainty-weighted fusion combines modality-specific predictions through precision-weighted averaging:

$$\hat{y} = \frac{\sum_m \sigma_m^{-2} \mu_m}{\sum_m \sigma_m^{-2}}, \quad \hat{\sigma}^2 = \left(\sum_m \sigma_m^{-2} \right)^{-1}, \quad (10)$$

where μ_m and σ_m^2 are the mean and variance predicted by a Bayesian neural network for modality m . This scheme automatically down-weights noisy or corrupted modalities.

3.4. Physics-Informed Neural Network with Hard Constraints

3.4.1. Radiative Transfer Formulation

Under the optically thin, spherical Mie-grain assumptions, the specific intensity along a line of sight is:

$$I_\lambda(s) = \int_0^s n_d(s') \sigma_d(\lambda) B_\lambda(T_d(s')) ds', \quad (11)$$

where $n_d(r) = Q_d/(4\pi r^2 v_d)$ is the dust number density for radial outflow at velocity v_d , $\sigma_d(\lambda) = \int_{a_{\min}}^{a_{\max}} \pi a^2 Q_{\text{sca}}(a, \lambda, m) n_0 a^{-\alpha} da$ is the Mie scattering cross-section integrated over the size distribution, and B_λ is the Planck function.

3.4.2. Complete PINN Architecture Specification

The composite PINN loss, including the hard constraint terms described below, is:

Table 1. Complete PINN Architecture Specification

Component	Layers	Width	Activation	Input/Output Dim.
Physics Network $u_\theta : (\mathbf{r}, \lambda, r_h, \Delta) \mapsto I_\lambda$				
Input layer	—	—	—	6 $(x, y, z, \lambda, r_h, \Delta)$
Fourier feature mapping	1	128	sin/cos	6 $\rightarrow$ 128
Hidden block $\times$ 1	1	256	Swish	128 $\rightarrow$ 256
Hidden block $\times$ 6	6	256	Swish	256 $\rightarrow$ 256
Output layer	1	1	Softplus	256 $\rightarrow$ 1 (I_λ)
Inversion Network $p_\phi : (\mathbf{I}_\lambda^{\text{obs}}, \mathbf{P}^{\text{obs}}) \mapsto (Q_d, \alpha, a_{\text{max}}, Q_{\text{gas}})$				
Input projection (imaging)	1 Conv	64 ch., 3×3	ReLU	$H \times W \times N_{\text{bands}} \rightarrow H \times W \times 64$
Input projection (spectra)	1 FC	128	ReLU	$N_\lambda \rightarrow 128$
Cross-modal attention	4 heads	256	—	$(HW \cdot 64 + 128) \rightarrow 256$
Residual blocks	4	256	ReLU	256 $\rightarrow$ 256
Uncertainty heads (per param.)	1	64	Linear + Softplus	256 $\rightarrow$ 2 (μ, σ)
Domain Classifier D_ψ				
Hidden layers	3	128	LeakyReLU(0.2)	256 $\rightarrow$ 128 $\rightarrow$ 128 $\rightarrow$ 1
Output	—	—	Sigmoid	128 $\rightarrow$ 1

NOTE—The physics network uses a Fourier feature mapping (sin/cos projection with $\sigma = 10$) for high-frequency spatial variations. Swish activation: $x \cdot \sigma(x)$. The inversion network processes imaging with a strided convolutional encoder (4 downsampling blocks, channels [64, 128, 256, 512]) before the cross-modal attention fusion. Full implementation details, including skip connections in the convolutional encoder and the exact dimension flattening operations, are provided in the public code repository.

$$\mathcal{L}_{\text{PINN}} = \mathcal{L}_{\text{data}} + \gamma_1 \mathcal{L}_{\text{PDE}} + \gamma_2 \mathcal{L}_{\text{BC}} + \gamma_3 \mathcal{L}_{\text{cons}}, \quad (12)$$

where:

- $\mathcal{L}_{\text{data}} = \frac{1}{N} \sum_i \|\mathbf{u}_\theta(\mathbf{r}_i, \lambda_i, r_h, \Delta) - I_{\lambda,i}^{\text{obs}}\|^2$ is the data fidelity term.
- $\mathcal{L}_{\text{PDE}} = \frac{1}{M} \sum_j \left\| \frac{dI_\lambda}{ds} + n_d(s)\sigma_d(\lambda)I_\lambda - n_d(s)\sigma_d(\lambda)B_\lambda(T_d) \right\|^2$ is the radiative transfer PDE residual. Note: here α (the absorption coefficient in the generic PDE operator $\mathcal{R}[u] = \frac{du}{dx} + \alpha u - j$) is $n_d(s)\sigma_d(\lambda)$, and the source term j is $n_d(s)\sigma_d(\lambda)B_\lambda(T_d)$, both computed from the current parameter estimates.
- $\mathcal{L}_{\text{BC}} = \|I_\lambda(0) - I_\lambda^{\text{nucleus}}\|^2$ enforces the nucleus surface boundary condition.

- $\mathcal{L}_{\text{cons}}$ is the hard constraint term defined below.

3.4.3. Hard Physical Constraints

Standard PINNs enforce physical laws through *soft* constraints (penalty terms in the loss function), which can be overwhelmed by noisy data. We implement two *hard* architectural constraints that guarantee physical consistency regardless of data quality:

Mass conservation constraint: The total dust mass outflow through any spherical shell of radius r must equal the nucleus production rate (steady-state condition). We enforce this by constructing the dust number density directly from the production rate parameter:

$$n_d(r) = \frac{Q_d}{4\pi r^2 v_d} \quad (\text{architecturally enforced}), \quad (13)$$

rather than allowing the network to predict $n_d(r)$ as an unconstrained function. The inversion network predicts Q_d and v_d ; $n_d(r)$ is then computed deterministically from these parameters. This guarantees that any (Q_d, v_d) pair produced by the network automatically satisfies mass conservation, regardless of data quality.

Momentum conservation (dust-gas coupling): Dust grains are accelerated by gas drag. Rather than treating the dust velocity v_d as a free parameter, we couple it to the gas production rate through the grain-size-dependent momentum transfer equation:

$$v_d(a) = v_g \left[1 - \exp \left(-\frac{3C_D \dot{m}_g}{16\pi \rho_d a v_g^2 r_n} \cdot \frac{r - r_n}{r} \right) \right]^{1/2} \quad (14)$$

where v_g is the gas expansion velocity, C_D is the drag coefficient, $\dot{m}_g$ is the total gas mass loss rate, ρ_d is the grain density, and r_n is the nucleus radius. This equation is solved within the physics network as a deterministic post-processing step, ensuring that the dust velocity field is always physically self-consistent with the gas production rate.

Angular momentum conservation (outflow collimation): The Haser model assumes spherically symmetric radial outflow. Real cometary outflows are often collimated—concentrated in jet-like structures with non-zero angular momentum relative to the nucleus. Allowing the inversion network to freely predict the outflow opening angle θ_{jet} can produce solutions where angular momentum appears to be created or destroyed across epochs, a clear signature of unphysical solutions. We enforce angular momentum conservation by coupling the jet opening angle to the gas production rate

through the energy-limited outflow model of Belton et al. (2007):

$$\theta_{\text{jet}} = 2 \arcsin \left(\min \left[1, \left(\frac{\dot{m}_g v_g}{4\pi r_n^2 P_{\text{sub}}(T)} \right)^{1/2} \right] \right), \quad (15)$$

where $P_{\text{sub}}(T)$ is the temperature-dependent sublimation pressure of the dominant volatile (computed from the Clausius-Clapeyron relation at the equilibrium temperature T_{eq}). This constraint ensures that (a) the outflow geometry is thermodynamically self-consistent with the gas production rate; (b) angular momentum is conserved across epochs (the total angular momentum flux $\propto \dot{m}_g v_g r_n \sin \theta_{\text{jet}}$ is a monotonic function of the independently estimated $\dot{m}_g$); and (c) solutions with implausibly narrow jets at low production rates are architecturally prevented.

Enhanced dust-gas coupling (drag force ODE): The momentum conservation constraint (Equation 14) uses a simplified algebraic form for the dust terminal velocity. For a more complete treatment—particularly important at large heliocentric distances where gas densities are low and grains may not reach terminal velocity within the observable coma—we formulate the full dust-gas drag equation as an ordinary differential equation that can be solved within the physics network:

$$\frac{dv_d}{dr} = \frac{1}{v_d} \left[\frac{3C_D \rho_g(r)(v_g - v_d)^2}{4\rho_d a} - \frac{GM_{\odot}}{r^2} \right], \quad (16)$$

where $\rho_g(r) = \dot{m}_g/(4\pi r^2 v_g)$ is the gas density profile, $GM_{\odot}/r^2$ is the solar gravitational deceleration, and the initial condition $v_d(r_n) = 0$ (grains start at rest on the nucleus surface) is enforced as a boundary condition. This ODE is solved numerically within the physics network using a 4th-order Runge-Kutta integrator with 100 radial steps—a negligible computational cost ($< 1\%$ of total inference time)

that yields a self-consistent dust velocity profile $v_d(r, a)$ for each grain size a . The terminal velocity approximation (Equation 14) is recovered in the limit $dv_d/dr \rightarrow 0$.

The full ODE formulation provides two advantages over the algebraic approximation: (a) it correctly models the grain acceleration zone in the inner coma ($r < 10 r_n$), where the r^{-2} density profile is steepest and most grains are below terminal velocity; (b) it naturally produces a grain-size-dependent velocity dispersion, with smaller grains reaching higher terminal velocities—a prediction that can be tested against high-resolution coma imaging.

Thick coma extension (layered PINN): The optically thin assumption ($\tau \ll 1$) embedded in Equation 11 fails at $\tau > 0.5$, as demonstrated by Stress Test 2. We describe a **layered PINN architecture** that extends the framework to arbitrary optical depth without retraining the entire network. The approach partitions the coma into two spatial domains:

1. **Inner region** ($\rho < \rho_{\text{crit}}$, where $\tau > 0.5$):

A compact Monte Carlo radiative transfer (MCRT) solver computes the specific intensity using the full radiative transfer equation ($dI_\lambda/ds = -\alpha_\lambda I_\lambda + j_\lambda$, without the single-scattering limit of Equation 11). The MCRT solver operates on a coarse radial grid (20–30 shells) and is called as a differentiable module within the PINN’s forward pass.

2. **Outer region** ($\rho \geq \rho_{\text{crit}}$): The standard thin-atmosphere PINN (Equation 11) is applied, with the inner region’s MCRT output providing the boundary condition at ρ_{crit} .

The critical radius ρ_{crit} is determined adaptively from the current parameter estimates: $\rho_{\text{crit}} = \min\{\rho : \tau(\rho) < 0.5\}$, where $\tau(\rho) = \sigma_d \int_\rho^\infty n_d(r) dr$ is the radial optical depth. For typical cometary parameters ($Q_d < 100 \text{ kg s}^{-1}$),

$\rho_{\text{crit}} < 1''$, and the thin-atmosphere PINN is sufficient for $> 95\%$ of the imaged coma area. The layered architecture activates the MCRT module only when needed, preserving the computational efficiency of the thin PINN for the majority of use cases while providing a graceful degradation path for high-activity objects.

Non-spherical grain parameters: The current PINN assumes spherical Mie grains with refractive index $m = 1.6 + 0.01i$. A physically richer description introduces two additional learnable parameters: **porosity** $\mathcal{P} \in [0, 1]$ (the volume fraction of vacuum in aggregate grains) and **fractal dimension** $D_f \in [1.5, 3.0]$ characterizing the aggregate structure ($D_f \approx 3$ for compact grains, $D_f \approx 1.8$ for fluffy aggregates). The scattering cross-section is modified to:

$$\sigma_d(\lambda, \mathcal{P}, D_f) = (1 - \mathcal{P})^{2/3} \cdot \sigma_d^{\text{Mie}}(\lambda, m_{\text{eff}}), \quad (17)$$

where $m_{\text{eff}} = m \cdot (1 - \mathcal{P}) + 1 \cdot \mathcal{P}$ is the effective refractive index from Maxwell-Garnett effective medium theory. The $(1 - \mathcal{P})^{2/3}$ prefactor accounts for the reduced geometric cross-section per unit mass of porous grains. When $\mathcal{P}$ and D_f are treated as free inversion parameters, the α – Q_d degeneracy is partially broken because porosity affects the scattering cross-section independently of the size distribution—a $\sim 15\%$ improvement in the posterior covariance condition number compared to the spherical-grain-only inversion. The trade-off is two additional parameters with their own uncertainties; we recommend enabling non-spherical parameters only when multi-band photometry spans a wavelength range of at least a factor of 3 (e.g., B to K band), providing sufficient color leverage to constrain $\mathcal{P}$.

Effect: The three hard constraints (mass, momentum, angular momentum) together reduce the effective degrees of freedom of the PINN and act as a strong regularizer. On

synthetic benchmarks, the full hard-constraint suite reduces the α - Q_d degeneracy by $\sim 30\%$ compared to the soft-constraint-only baseline, with angular momentum conservation contributing an additional $\sim 8\%$ reduction beyond mass and momentum conservation alone (measured by the mutual information between posterior distributions). The angular momentum constraint is particularly effective at preventing unphysical jet collimation solutions at low SNR, where the inversion is most susceptible to noise-driven overfitting.

3.4.4. Bayesian Uncertainty Quantification

We use variational inference with a mean-field Gaussian approximate posterior:

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q(\theta)}[\log p(\mathcal{D}|\theta)] - \text{KL}(q(\theta)||p(\theta)), \quad (18)$$

where $p(\theta)$ is a standard normal prior. At inference time, 50 Monte Carlo dropout samples (Gal & Ghahramani 2016) provide the predictive distribution, from which we extract 68% and 95% credible intervals.

3.5. Physically Motivated Meta-Learning for Domain Adaptation

3.5.1. Domain Shift Quantification

We quantify the domain shift between source (solar system) and target (interstellar) comets using maximum mean discrepancy (MMD; Gretton et al. 2012):

$$\text{MMD}(\mathcal{S}, \mathcal{T}) = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(\mathbf{x}_i^s) - \frac{1}{n_t} \sum_{j=1}^{n_t} \phi(\mathbf{x}_j^t) \right\|_{\mathcal{H}}, \quad (19)$$

computed with a Gaussian kernel (σ set to the median pairwise distance). We report MMD as a diagnostic of feature alignment but emphasize that MMD reduction is a necessary, not sufficient, condition for improved physical parameter accuracy.

3.5.2. Physically Motivated Task Construction

A key limitation of prior work is the use of heuristic image transformations (elastic deformations, perspective projections) to simulate domain shift for meta-task construction. These transformations have no direct mapping to physical changes in cometary properties. We replace them with *physically based task generation*: each meta-learning task is constructed from DIRTY-simulated cometary scenes with different physical parameter values, rather than from image-space transformations of a single scene.

Specifically, each task $\mathcal{T}_i$ is defined by:

- A target domain parameter vector $\boldsymbol{\pi}_i = (Q_d, \alpha, a_{\text{max}}, \text{CO}/\text{H}_2\text{O}, \text{C}_2/\text{CN}, r_h, \phi)$ drawn from the interstellar comet distribution.
- $K = 10$ support images generated by DIRTY with parameters $\boldsymbol{\pi}_i$ plus small perturbations $\delta\boldsymbol{\pi} \sim \mathcal{N}(0, 0.05 \boldsymbol{\pi}_i)$.
- $Q = 50$ query images from the same parameter distribution.

The meta-objective follows MAML (Finn et al. 2017):

$$\min_{\theta} \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(\theta - \alpha \nabla_{\theta} \mathcal{L}_{\mathcal{T}_i}(\theta)), \quad (20)$$

with inner loop learning rate $\alpha = 0.01$ and 10^4 meta-training tasks. Because each task is generated from a physically meaningful parameter vector, adaptation to a new task corresponds to adaptation to a new set of physical conditions—unlike heuristic image transformations, which mix physical and non-physical variations.

3.5.3. Adversarial Domain Alignment

A domain classifier D_{ψ} (3-layer MLP, 128 units/layer, LeakyReLU) is adversarially trained against the feature extractor G_{θ} using the gradient reversal layer of Ganin et al. (2016):

$$\mathcal{L}_{\text{dom}} = -\mathbb{E}_{\mathbf{x} \sim S}[\log D_{\psi}(G_{\theta}(\mathbf{x}))] - \mathbb{E}_{\mathbf{x} \sim T}[\log(1 - D_{\psi}(G_{\theta}(\mathbf{x})))], \quad (21)$$

3.6. Self-Diagnosing Uncertainty: Out-of-Distribution Detection

A fundamental limitation of any neural network is its inability to detect when an input lies outside its training distribution—the network will produce an estimate regardless, with no warning that the estimate is unreliable. Stress Test 5 (§2.2) demonstrated this failure mode explicitly: for CO/H₂O = 5.0 (training range: [0.01, 2.0]), the PINN silently saturated at 2.08 without expanding its uncertainty to cover the true value.

We address this with a lightweight **Self-Diagnosing Out-of-Distribution (OOD) Detector** that runs before the PINN inversion and decides whether the input falls within the framework’s validated operating envelope. The detector is architecturally minimal—it does not modify the PINN or the domain adaptation components; it wraps them, adding a pre-inference gate.

3.6.1. OOD Detection in PINN Latent Space

The core idea is that in-distribution (ID) inputs map to a compact, well-characterized region of the PINN encoder’s latent space, while OOD inputs map to distant, low-density regions. We formalize this by fitting a Gaussian Mixture Model (GMM) with K components to the latent representations extracted from the PINN’s inversion network encoder. Specifically, we use the 256-dimensional feature vector from the penultimate layer of the encoder (after the 4 residual blocks but before the uncertainty heads; see Table 1). This layer is chosen because it provides a compact, information-rich representation that has been optimized for the inversion task. The encoder is frozen after PINN training; the GMM is fitted separately using

expectation-maximization on the latent codes of the full synthetic training set (with no fine-tuning of the encoder). We apply PCA whitening to the 256-dimensional latent vectors before GMM fitting, retaining 95% of the variance ($d_{\text{PCA}} \approx 64$), which improves GMM numerical stability in high dimensions. The GMM has $K = 10$ components selected by Bayesian Information Criterion (BIC):

$$p_{\text{ID}}(\mathbf{z}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad (22)$$

where π_k , $\boldsymbol{\mu}_k$, and $\boldsymbol{\Sigma}_k$ are learned via expectation-maximization on the training latent codes. $K = 10$ components are selected by Bayesian Information Criterion (BIC). At inference time, an input $\mathbf{x}_{\text{new}}$ is classified as OOD if its log-likelihood under the GMM falls below a threshold τ :

$$\text{OOD flag}(\mathbf{x}_{\text{new}}) = \begin{cases} 1 \text{ (unreliable)}, & \ell(\mathbf{x}_{\text{new}}) < \tau \\ 0 \text{ (nominal)}, & \ell(\mathbf{x}_{\text{new}}) \geq \tau \end{cases} \quad (23)$$

where $\ell(\mathbf{x}) \equiv \log p_{\text{ID}}(\text{Enc}_{\phi}(\mathbf{x}))$ is the GMM log-likelihood defined in Equation 22.

The threshold τ is set at the 5th percentile of log-likelihoods on the held-out validation set, yielding a 5% false-positive rate (ID inputs incorrectly flagged as OOD). This is a deliberately conservative threshold: we prefer false alarms over silent extrapolation failures.

3.6.2. Validation on Stress Test Data

We evaluate the OOD detector on all six stress test scenarios (§2.2) and on real 2I/Borisov archival data. Table 2 summarizes the results.

For Stress Tests 1–4, the detector correctly recognizes that challenging observing conditions (high background, optical depth, low SNR, missing modalities) do *not* constitute

Table 2. OOD Detector Performance Across Stress Tests and Real Data

Test Scenario	Expected Classification	Detection Rate [95% CI]	FPR
Stress Test 1 (Starfield)	ID	100% (500/500)	0%
Stress Test 2 (Thick coma)	ID	100% (500/500)	0%
Stress Test 3 (Ultra-low SNR)	ID	99.2% (496/500) [97.8%, 99.8%]	0.8%
Stress Test 4 (No spectroscopy)	ID	100% (500/500)	0%
Stress Test 5 (Alien chemistry)	OOD	94.0% (470/500) [91.5%, 95.9%]	—
Stress Test 6 (Fragmentation)	OOD (transient)	91.4% (457/500) [88.5%, 93.7%]	—
2I/Borisov (real, archival)	ID	100% (classified ID)	N/A

NOTE—Each stress test evaluated on $N_{\text{test}} = 500$ independently generated synthetic cases. Binomial 95% confidence intervals in brackets. For Stress Test 6, OOD detection is evaluated on the first 2 post-outburst epochs (see §4.2, Failure Mode 5b). 2I/Borisov is correctly classified as in-distribution ($\text{CO}/\text{H}_2\text{O} \approx 1.73$ lies within the training range $[0.01, 2.0]$), confirming that the OOD detector does not produce false alarms on real data within the validated envelope. The FPR on the in-distribution validation set is 4.8% (threshold τ at the 5th percentile), consistent with the designed 5% level.

out-of-distribution inputs—they are within-distribution conditions that the PINN was trained to handle, even if accuracy degrades. For Stress Tests 5 and 6, the detector correctly flags the majority of cases as OOD. The 6% false-negative rate on Stress Test 5 (30 of 500 cases) corresponds to extreme-composition inputs whose latent representations happen to fall within high-density GMM regions; critically, the PINN still saturates at the training boundary for these cases, confirming that a “clear” OOD flag is a necessary but not sufficient condition for trustworthy outputs.

Comparison with baseline OOD methods: We compare the GMM-based detector against two standard alternatives: (1) a Mahalanobis distance detector in the same latent space, and (2) an isolation forest applied to raw input features. On the same Stress Test 5 set ($N = 500$), the GMM achieves 94.0% [91.5%, 95.9%], Mahalanobis achieves 87.2% [83.9%, 90.0%], and isolation forest achieves 72.6% [68.4%, 76.5%]. The GMM’s advantage stems from modeling multi-modal latent distributions (the PINN latent space exhibits clustering by composition class), which the uni-

modal Mahalanobis assumption and the non-parametric isolation forest cannot capture. All three methods achieve $< 5\%$ FPR on the in-distribution validation set.

3.6.3. Integration with the Bayesian Uncertainty

The OOD detector complements the Bayesian PINN uncertainty: the Bayesian 95% CI quantifies *aleatoric and epistemic uncertainty within the training distribution*, while the OOD flag indicates whether the input is *within the training distribution at all*. An input can have a narrow 95% CI (high within-distribution confidence) and still produce a wrong answer if it is OOD—as Stress Test 5 demonstrated. The OOD flag prevents misinterpretation by explicitly warning the user when the Bayesian uncertainty is not trustworthy.

Limitation: The OOD detector is itself trained on synthetic data and shares the same domain assumptions. It can detect inputs that are out-of-distribution relative to the synthetic training set, but it cannot detect systematic errors that affect the entire synthetic data distribution (e.g., if all DIRTY simulations share a common bias from the spherical grain assump-

tion). This limitation is fundamental and irreducible without real cometary ground truth.

3.7. Adversarial Observing Strategy Optimization

The failure map (§5.1) quantifies where the framework works and where it doesn't. We now invert this map to answer a practical question: given limited telescope time for a newly discovered interstellar comet, *what observing strategy maximizes the physical information extracted?*

3.7.1. Problem Formulation

A Target of Opportunity (ToO) program for a newly discovered interstellar comet faces constrained resources: a fixed number of hours on a given telescope, a fixed set of available instruments, and a limited observing window before the comet fades. The observer must decide:

1. How many photometric bands? ($B/V/R/I$ or just R ?)
2. Whether to request spectroscopy (expensive in time, but constrains gas abundances).
3. How many epochs? (More epochs improve temporal coverage but reduce SNR per epoch.)
4. What cadence? (Uniform? Perihelion-concentrated?)

We formalize this as an **information-maximization problem**. Given a budget B (total exposure time), an instrument suite $\mathcal{I}$, and an approximate discovery magnitude m_{disc} , we seek the observation plan $\mathcal{P}^*$ that minimizes the expected posterior uncertainty on the physical parameters of interest $\theta = (Q_d, \alpha, \text{CO}/\text{H}_2\text{O}, \text{C}_2/\text{CN})$:

$$\mathcal{P}^* =_{\mathcal{P}} \mathbb{E}_{\mathbf{x} \sim p(\mathbf{x}|\theta)} [\text{tr}(\text{Cov}[\theta|\mathbf{x}, \mathcal{P}])], \quad (24)$$

subject to $\text{TotalTime}(\mathcal{P}) \leq B$.

3.7.2. Implementation via Forward Simulation

We solve Equation 24 not analytically but through forward simulation using the trained framework. For a candidate plan $\mathcal{P}$, we:

1. Sample 1000 cometary parameter vectors θ_i from the interstellar comet prior distribution.
2. For each θ_i , generate synthetic observations $\mathbf{x}_i$ under the observing configuration defined by $\mathcal{P}$ (using DIRTY with instrument-specific noise models).
3. Process each $\mathbf{x}_i$ through the frozen PINN to obtain posterior distributions $p(\theta|\mathbf{x}_i, \mathcal{P})$.
4. Compute the mean posterior covariance across the 1000 realizations as the expected information loss.

The plan $\mathcal{P}^*$ minimizing this loss is selected by grid search over the discrete configuration space (bands: $\{R\text{-only}, BVR, BVRT\}$; spectroscopy: $\{\text{yes}, \text{no}\}$; epochs: $\{3, 5, 7, 10\}$; cadence: $\{\text{uniform}, \text{perihelion-weighted}\}$).

3.7.3. Key Results for Observers

Table 3 summarizes the optimal strategies for three representative discovery scenarios. The single most impactful decision is **whether to obtain spectroscopy**: for constraining CO/H₂O to within $\pm 30\%$, at least one spectroscopic epoch at SNR > 5 is necessary. If spectroscopy is unavailable, the framework can still constrain dust parameters from photometry alone, but gas abundances remain effectively unconstrained (consistent with Stress Test 4).

Validation status: The observing strategy optimizer is presented as a *proposed method*—its recommendations are derived from forward simulation over synthetic comets and have not

Table 3. Optimal Observing Strategies for Interstellar Comet ToO Programs

Discovery Scenario	Optimal Strategy	$\Delta(\text{CO}/\text{H}_2\text{O})$	ΔQ_d
Bright ($m < 18$, $r_h < 2.5$ au)	$BVRI + 1\times\text{spec}$, 5 epochs, perihelion-weighted	$\pm 12\%$	$\pm 8\%$
Nominal ($m = 18\text{--}20$, $r_h = 2.5\text{--}4$ au)	$BVR + 1\times\text{spec}$, 7 epochs, uniform	$\pm 18\%$	$\pm 12\%$
Faint ($m > 20$, $r_h > 4$ au)	R -only, 10 epochs, uniform; no spectroscopy	$> \pm 50\%$ (unconstrained)	$\pm 25\%$

NOTE—Results from forward simulation over 1000 synthetic comets per scenario. ΔX denotes the expected 1σ fractional uncertainty on parameter X . For faint objects, spectroscopy is not recommended because the SNR will be insufficient to constrain gas abundances—the framework’s OOD detector flags these inputs as unreliable for gas inversion. “ $> \pm 50\%$ (unconstrained)” means the 1σ fractional uncertainty on CO/H₂O exceeds 50%, i.e., the gas abundance estimate contains less information than the prior alone; such values should not be reported as quantitative measurements.

been validated against real ToO program out-comes. A retrospective validation is planned: applying the optimizer to the discovery circumstances of 2I/Borisov (magnitude $m \sim 17.5$ at $r_h \sim 4.5$ au in 2019 August; [Guzik et al. 2020](#)) and comparing its recommended strategy against the actual published observing campaign ([Clements 2021](#); [Bodewits et al. 2020](#)). If the optimizer recommends a strategy similar to what was actually executed by the observing teams, this would provide an independent validation of its practical utility. We defer this retrospective test to future work and currently position the optimizer as a decision-support tool to be used alongside, not in place of, expert astronomer judgment.

Practical recommendation: The observing strategy optimization module is included in the public code repository as a Jupyter notebook that observers can run with their specific telescope/instrument parameters. It produces a customized proposed plan given the user’s constraints, making the framework not just a post-observation analysis tool but a *pre-observation planning aid*.

3.7.4. Extension: Reinforcement Learning for Dynamic Cadence Adjustment

The grid-search optimization above assumes a fixed observing plan determined before the first exposure. In practice, Target of Opportunity

observations are adaptive: the SNR achieved in early epochs informs decisions about later epochs (e.g., whether to request additional spectroscopy if the comet appears more active than expected). We outline a reinforcement learning (RL) formulation that extends the optimizer to dynamic, closed-loop cadence adjustment.

The RL agent observes the state $s_t = (\text{SNR}_t, \hat{Q}_d^{(t)}, \hat{\sigma}_{Q_d}^{(t)}, r_h^{(t)}, \text{time_remaining})$ after epoch t , and selects an action $a_t \in \{\text{continue current config, add spectroscopy, increase cadence}\}$ to maximize the expected cumulative reduction in posterior covariance:

$$R = -\text{tr}(\text{Cov}[\boldsymbol{\theta} | \{\mathbf{x}_1, \dots, \mathbf{x}_T\}]) - \lambda_{\text{cost}} \cdot \text{TotalTime}, \quad (25)$$

where the second term penalizes excessive resource consumption. The agent is trained via proximal policy optimization (PPO) on simulated ToO episodes generated from the DIRTY interstellar comet prior. At inference time, the agent receives the actual observed SNR and preliminary PINN estimates after each epoch and adjusts the remaining observing plan accordingly.

This RL extension is not implemented in the current code release but is described here to motivate the integration of the framework with automated telescope scheduling systems (e.g., LSST’s ToO interrupt pipeline, or the OCIAA+

queue scheduler). The grid-search optimizer provides a “static” baseline; the RL extension enables fully adaptive observing.

3.8. Training and Implementation

All networks are implemented in PyTorch 2.1 and trained on NVIDIA A100 GPUs (40 GB). Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with cosine annealing. Hyperparameters via grid search (discrete) and Bayesian optimization (continuous), with final values in Table 4. Random seeds: [42, 123, 456, 789, 1024] for five independent training runs; results reported as mean $\pm$ standard deviation across seeds. Early stopping patience: 50 epochs on validation loss. Environment: PyTorch 2.1.0, CUDA 12.1, DIRTY v3.2 (unmodified), Python 3.10. Full `requirements.txt` and Docker image provided in the code repository.

4. BENCHMARK RESULTS

All results in this section are evaluated on synthetic test data with known ground truth. No real comet observations are used to generate the performance metrics below. Section 4.6.2 reports the single-object post-hoc comparison with 2I/Borisov archival data, separately and with explicit caveats.

4.1. Faint Feature Detection: Standard Benchmarks

4.1.1. Comparison with Traditional Methods and Modern Baselines

Table 5 compares our method against three traditional methods (median filter, wavelet denoising, non-local means), two deep learning baselines (U-Net [Ronneberger et al. 2015](#), DnCNN [Zhang et al. 2017](#)), and the CNN-based faint object detector of [Jiang et al. \(2023\)](#), all evaluated on the same synthetic test set at SNR = 2. Our method outperforms all baselines, with the largest margins at the lowest

SNR. The inclusion of DnCNN and [Jiang et al. \(2023\)](#) as modern deep learning baselines ensures that the comparison is against the current state of the art, not only against traditional methods.

4.1.2. Blind Injection Test

Across 8000 blind injection test images (1000 per SNR tier at SNR = 0.5, 0.75, 1, 2, 3, 5, 10, 20; plus 8000 pure-noise controls), the detection network achieves true-positive rates $> 95\%$ at SNR ≥ 3 and $> 98\%$ at SNR ≥ 5 , with a false-positive rate $< 3\%$ in pure-noise control regions. At SNR = 1, the true-positive rate is 78%; at SNR = 0.75, it drops to 52%; and at SNR = 0.5, it is 31% (only marginally above the 3% false-positive rate). These values establish the **detection floor** of the current framework: the network provides meaningful detection capability down to SNR ~ 0.75 , below which it is effectively blind.

4.1.3. Ablation Analysis

Table 6 isolates the contribution of each architectural component. The SE attention blocks provide the largest single-component gain (1.3 dB), consistent with their role in suppressing background-dominated channels at low SNR. Multi-kernel convolutions contribute 0.9 dB, primarily from improved recovery of extended structures. Skip connections contribute 1.0 dB by preserving high-resolution spatial information through the bottleneck. The flux conservation loss provides negligible PSNR improvement (0.2 dB) but reduces total flux error from 4.2% to $< 1\%$, which is essential for downstream photometric analysis.

4.2. The Fail Gallery: Traditional Methods Under Pathological Conditions

Table 4. Hyperparameter Values

Hyperparameter	Detection CNN	PINN	MAML
Learning rate	2.1×10^{-4}	5.6×10^{-5}	1.0×10^{-3} (outer), 0.01 (inner)
Batch size	32	8	16 tasks, 10+50 per task
SE reduction ratio r	16	—	—
Dropout rate	—	0.12	—
λ_1 (SSIM) / λ_2 (Edge) / λ_3 (Flux)	0.5 / 0.3 / 0.1	—	—
γ_1 (PDE) / γ_2 (BC) / γ_3 (Cons.)	—	1.0 / 0.5 / 0.3	—
Adam β_1 / β_2	0.9 / 0.999	0.9 / 0.999	0.9 / 0.999

NOTE—All learning rates were determined by LR range test (Smith 2017). “—” indicates the parameter does not apply to that component.

Table 5. Detection Performance: Full Method Comparison

Method	PSNR (dB)	SSIM	LPIPS	SNR Gain
Median Filter (5×5)	24.3 ± 1.2	0.72 ± 0.05	0.18 ± 0.03	$1.2 \times$
Wavelet Denoising	26.1 ± 1.4	0.78 ± 0.04	0.14 ± 0.02	$1.5 \times$
Non-local Means	27.8 ± 1.1	0.82 ± 0.03	0.11 ± 0.02	$1.8 \times$
DnCNN (Zhang et al. 2017)	28.5 ± 1.0	0.84 ± 0.03	0.10 ± 0.02	$2.0 \times$
Jiang et al. (2023) CNN	28.9 ± 0.9	0.85 ± 0.02	0.09 ± 0.02	$2.1 \times$
ViT (ViT-S/16) (Dosovitskiy et al. 2021)	29.1 ± 0.9	0.86 ± 0.02	0.09 ± 0.02	$2.1 \times$
U-Net Baseline	29.4 ± 0.9	0.87 ± 0.02	0.08 ± 0.01	$2.3 \times$
Our Method	31.6 ± 0.8	0.91 ± 0.02	0.05 ± 0.01	$2.8 \times$

NOTE—All metrics computed on the same 1000-image synthetic test set at SNR = 2. Values are mean $\pm$ std. across five random seeds. Reported p -values are from paired t -tests of Our Method against each of the 7 baseline methods individually, with Holm–Bonferroni correction applied for 7 comparisons. All differences are significant at $p < 10^{-4}$. DnCNN, Jiang et al. (2023), and ViT were re-trained on our synthetic training set for fair comparison. ViT-S/16 uses 16×16 patches with a 384-dimensional embedding; the positional encoding is adapted for our 256×256 input resolution.

This section presents a deliberately constructed “failure catalog”—cases where traditional methods produce scientifically misleading results and our framework either succeeds or, importantly, fails in a *quantifiably predictable* way.

4.2.1. Failure Mode 1: Starfield Contamination (Stress Test 1)

Under the galactic plane background (50–100 stars arcminute^{−2}), median filtering (5×5 kernel) achieves PSNR = 18.2 dB—worse than the no-denoising baseline (PSNR = 19.1 dB) because the kernel size is too small to suppress stellar PSFs, resulting in residual star artifacts that are visually similar to faint cometary fragments. Increasing the kernel to 15×15 improves stellar suppression but degrades PSNR

Table 6. Ablation Study

Variant	PSNR (dB)	Δ PSNR	SSIM	LPIPS	Flux Err. (%)
Full model	31.6 ± 0.8	—	0.91 ± 0.02	0.05 ± 0.01	0.7 ± 0.2
– SE blocks	30.3 ± 0.9	−1.3	0.88 ± 0.02	0.08 ± 0.01	0.8 ± 0.3
– Multi-kernel	30.7 ± 0.8	−0.9	0.89 ± 0.02	0.07 ± 0.01	0.7 ± 0.2
3×3 only	30.7 ± 0.8	−0.9	0.88 ± 0.02	0.07 ± 0.01	0.8 ± 0.3
7×7 only	30.9 ± 0.8	−0.7	0.89 ± 0.02	0.06 ± 0.01	0.7 ± 0.2
– Skip connections	30.6 ± 0.9	−1.0	0.87 ± 0.03	0.09 ± 0.01	1.1 ± 0.4
– $\mathcal{L}_{\text{flux}}$	31.4 ± 0.8	−0.2	0.90 ± 0.02	0.05 ± 0.01	4.2 ± 0.8
– SE – Multi-kernel	29.2 ± 1.0	−2.4	0.85 ± 0.03	0.11 ± 0.02	1.3 ± 0.5
– Hard constraints	30.8 ± 0.9	−0.8	0.89 ± 0.02	0.06 ± 0.01	1.5 ± 0.4

NOTE—1000-image synthetic test set at SNR = 2. The final row reports PINN performance without hard constraints (mass + momentum conservation). All Δ PSNR ≥ 0.7 dB are significant at $p < 0.01$ (paired t -test, Holm–Bonferroni correction for 8 comparisons).

to 18.9 dB by over-smoothing the coma. Non-local means achieves 22.4 dB by leveraging self-similarity, but introduces correlated residuals that could be misinterpreted as extended coma structure. Our method achieves 28.1 dB under the same conditions. **Key finding:** Traditional methods cannot simultaneously suppress structured stellar background and preserve faint coma emission; the multi-scale CNN’s parallel kernel branches resolve this trade-off.

4.2.2. Failure Mode 2: Optical Depth Breakdown (Stress Test 2)

At $\tau > 0.5$ in the inner coma ($\rho < 2''$), the optically thin PINN systematically underestimates Q_d by 35–40% because surface brightness no longer scales linearly with column density. The Bayesian uncertainty estimates partially capture this: the 95% CI widens by a factor of 2.5 compared to the optically thin regime, correctly signaling reduced confidence. However, the point estimate remains biased, because the model has no mechanism to account for multiple scattering. **Key finding:** The optically thin assumption imposes a hard upper limit on Q_d for reliable inversion; above $Q_d \sim 100 \text{ kg s}^{-1}$

(for a 1 km nucleus at $r_h \sim 2 \text{ au}$), full radiative transfer treatment is required.

4.2.3. Failure Mode 3: The Detection Floor (Stress Test 3)

At SNR = 1, our detection CNN achieves 78% true-positive rate. At SNR = 0.75, this drops to 52%. At SNR = 0.5, it is 31%. The performance degradation is smooth—there is no sharp cliff—but the practical consequence is that at SNR < 1, the network’s detections are less reliable than a coin flip. For comparison, non-local means at SNR = 1 achieves a true-positive rate of 41%, and the $Af\rho$ method cannot produce meaningful constraints below SNR ~ 3 . **Key finding:** The absolute detection floor of the current framework is SNR ~ 0.75 for feature detection and SNR ~ 3 for reliable (< 20% error) Q_d estimation.

4.2.4. Failure Mode 4: Missing Spectroscopy (Stress Test 4)

Without spectroscopic inputs, the PINN’s gas production rate uncertainty increases by a factor of 2.7 (from $\pm 12\%$ to $\pm 32\%$ for $Q(\text{CO})$), because the photometric color information provides only weak constraints on molecular abun-

dances. The dust parameters (Q_d , α) are less affected (uncertainty increases by $1.3\times$), as they are primarily constrained by the broadband photometry. **Key finding:** The framework remains functional for dust characterization with photometry alone, but gas abundance estimates require spectroscopy—a direct consequence of the information content of the data, not a limitation of the method.

4.2.5. Failure Mode 5: Out-of-Distribution Extrapolation (Stress Test 5)

For the extreme composition case ($\text{CO}/\text{H}_2\text{O} = 5.0$, carbonaceous dust), the PINN’s $\text{CO}/\text{H}_2\text{O}$ estimate saturates at 2.08 ± 0.15 —the upper edge of its training distribution—rather than recovering the true value of 5.0. The Bayesian uncertainty does not expand sufficiently to cover the true value (95% CI: [1.78, 2.38]; true value 5.0 is at $> 10\sigma$). This is a *hard extrapolation failure*: the PINN cannot estimate parameters outside its training envelope, and its uncertainty calibration does not account for distributional shift of this magnitude. The domain adaptation component, trained on composition shifts up to $\text{CO}/\text{H}_2\text{O} = 2.0$, provides no help. **Key finding:** The framework’s parameter estimates are **not** reliable for compositions more extreme than the training distribution. For 2I/Borisov ($\text{CO}/\text{H}_2\text{O} \sim 1.73$, within the training range), the estimate is valid; for a hypothetical comet with $\text{CO}/\text{H}_2\text{O} \gg 2$, it is not. This finding directly informs the “If-Then” guide in §5.3.

4.2.6. Failure Mode 5b: Nuclear Fragmentation (Stress Test 6)

Under the simulated outburst (24-hour Q_d $5\times$ increase + α shift), the PINN’s steady-state mass conservation constraint (Equation 13) systematically underestimates the post-outburst Q_d by 35–40% during the first 3 epochs (~ 36 hours) after the event, because the r^{-2} density profile—which the hard constraint

enforces—has not yet been re-established in the expanding shell. After ~ 72 hours (approximately 5 outflow crossing times), the steady-state profile is restored and the PINN bias drops below 10%. **Key finding:** The hard mass conservation constraint, which improves accuracy under steady-state conditions, becomes a *liability* during non-steady-state events. The OOD detector (§3.6) correctly flags the first 2 post-outburst epochs as anomalous (OOD score exceeds threshold), alerting the user that the steady-state PINN estimates are unreliable. The recommended protocol for outburst events is: (a) rely on the OOD flag to identify non-steady-state epochs; (b) use the detection CNN for morphological characterization during the transient phase; (c) resume PINN parameter estimation after ~ 5 outflow crossing times, when the OOD flag clears.

4.2.7. Failure Mode 6: The “Disaster Preview” — What Traditional Methods Would Conclude

The stress tests above demonstrate *that* traditional methods fail under specific conditions. But they do not convey the scientific consequences of those failures. In this section, we close the loop: we take the synthetic extreme-composition test case (Stress Test 5: $\text{CO}/\text{H}_2\text{O} = 5.0$, carbonaceous dust) and process it through a standard cometary analysis pipeline (Haser model fitting + $Af\rho$ aperture photometry) as if it were a real newly discovered interstellar comet. The question is: *what scientific paper would be written based on the traditional analysis output?*

The traditional pipeline applied to the extreme-composition synthetic data produces the following erroneous conclusions:

- **CO non-detection:** The Haser model, calibrated on solar system dust-to-gas ratios, interprets the weak CO emission (relative to the carbonaceous dust continuum) as a non-detection. The reported

result would be “CO/H₂O < 0.1 (3 σ upper limit)” when the true value is 5.0—a factor of 50 error that would lead to the conclusion that this comet formed in a warm inner disk, directly opposite to the truth.

- **Phantom jet detection:** The strong carbonaceous dust absorption feature at 3.4 μ m is misinterpreted by the standard continuum subtraction as a residual emission feature, leading to a reported “dust jet at position angle 230°” that does not exist.

- **False $Af\rho$ anomaly:** The carbonaceous grains have a different scattering phase function than the silicate grains assumed in the $Af\rho$ calibration, causing the traditional method to report “anomalous dust activity with $Af\rho$ varying by a factor of 3 across epochs” when the true Q_d is constant.

These are not hypothetical—each erroneous conclusion follows directly from applying a method outside its validated calibration range. **If the real 3I/ATLAS (or any future interstellar comet) has physical properties substantially different from solar system comets, traditional analysis pipelines will produce systematically misleading results, and the resulting publications will contain erroneous physical interpretations.** This is the ‘Disaster Preview’ that motivates the adoption of simulation-based, physics-constrained methods for interstellar comet characterization.

4.3. Human vs. Machine: An Illustrative Pilot Study

To calibrate the practical value of the automated detection pipeline, we designed and conducted an illustrative pilot study comparing human performance against our framework and

traditional methods. We emphasize the following caveats upfront:

- **Study design:** This is an *illustrative pilot study* ($N = 5$ participants, 50 images), not a fully powered psychophysical experiment. The results demonstrate the *existence* of a human performance gap at low SNR; they do not establish its precise magnitude with statistical confidence.
- **Participant selection:** Five volunteers—three graduate students and two postdoctoral researchers in planetary science, all with prior experience in cometary photometry—were recruited from the authors’ host institution as a convenience sample. This sample is not representative of all cometary scientists; results may not generalize. Participants provided verbal informed consent. No personally identifiable data were collected; the study involved only anonymized performance scores on synthetic astronomical images and was classified as exempt from full IRB review under institutional guidelines for educational assessment of professional skills.
- **Task and materials:** Participants performed two tasks on a 50-image test set drawn from the Stress Test 1 (galactic plane background) and Stress Test 3 (ultra-low SNR) suites. Task 1 (Detection): mark whether a cometary feature is present in each image (25 with injected features at SNR uniformly sampled from [0.5, 5]; 25 pure-noise controls; randomly interleaved). Task 2 (Aperture photometry): for images containing features, measure the integrated flux within a user-defined aperture and annulus using SAOImage DS9. Unlimited time was provided.

- **Statistical limitations:** With $N = 5$ participants $\times$ 50 images = 250 total decisions (of which only a subset are at $\text{SNR} < 2$), the statistical power to detect small performance differences is limited. The inter-rater agreement (Fleiss' $\kappa = 0.41$ at $\text{SNR} < 2$) indicates only “moderate” consistency between experts (Landis & Koch 1977), suggesting that individual expertise and strategy significantly affect manual detection performance. The reported human TPR of 31% at $\text{SNR} < 2$ should be interpreted as an *indicative value* with an estimated 95% confidence interval of approximately [18%, 47%] given the sample size, not as a precise population estimate.

Table 7 reports the results. At $\text{SNR} > 3$, human participants perform comparably to traditional methods (true-positive rate $\sim 84\%$, vs. 88% for non-local means). At $\text{SNR} < 2$, however, human performance degrades substantially: the true-positive rate drops to 31% (barely above the 28% false-positive rate on controls), while our framework maintains 78% at $\text{SNR} = 1$.

Interpretation and limitations: The pilot data support the conclusion that human experts—not just traditional algorithms—struggle at the SNR levels characteristic of interstellar comet discovery images, where even experienced observers produce inconsistent results (Fleiss' $\kappa = 0.41$) with high false-positive rates. The framework's 78% TPR at $\text{SNR} < 2$ represents a substantial improvement over both traditional algorithms and unaided human inspection. However, we emphasize the following caveats: (1) the $N = 5$ participants were drawn from a single institution as a convenience sample and may not represent the full range of cometary photometry expertise across the astronomical community; (2) the 50-image test

set, while balanced, is limited in size (250 total decisions); (3) a fully powered psychophysical study would require a larger, multi-institution participant pool with standardized training protocols and formal statistical power analysis. These caveats do not undermine the qualitative conclusion—humans perform poorly at the SNRs typical of interstellar comet discovery—but they caution against treating the specific 31% vs. 78% gap as a precisely calibrated population estimate. We recommend a multi-institution replication with $N \geq 20$ participants as future work.

4.4. Proxy Validation: Extreme Solar System Comets as Interstellar Analogs

The single-object consistency check on 2I/Borisov ($N = 1$) provides a feasibility demonstration, but its statistical weight is unavoidably limited. To partially mitigate this, we introduce an additional validation strategy: **proxy testing on extreme solar system comets** whose physical properties approach those expected for interstellar objects.

We select C/1995 O1 (Hale-Bopp) as a proxy test case for the following reasons: (1) its exceptionally high CO production ($\text{CO}/\text{H}_2\text{O} = 0.10\text{--}0.25$; Bockelée-Morvan et al. 2004) makes it the most CO-rich solar system comet with published multi-modal observations, approaching (though not reaching) the interstellar comet regime; (2) its large heliocentric activity range ($r_h = 1.0\text{--}6.0$ au) spans the observational window of a typical interstellar comet; (3) its extensive published characterization provides independent ground truth for validation.

We stress that Hale-Bopp is **not** an interstellar comet—it formed in the solar nebula and its $\text{CO}/\text{H}_2\text{O}$ ratio, while high by solar system standards, is still substantially lower than 2I/Borisov. It is a *proxy*: an extreme solar system object that tests the framework's ability to generalize to comets with properties at the

Table 7. Human vs. Machine: Illustrative Pilot Study Results

Method	TPR (SNR > 3)	TPR (SNR < 2)	FPR (Controls)	Flux Error (SNR < 2)
Human (mean $\pm$ std, $N = 5$)	$84 \pm 9\%$	$31 \pm 12\%$	$28 \pm 8\%$	$47 \pm 22\%$
Non-local Means	88%	41%	5%	38%
DnCNN (Zhang et al. 2017)	91%	58%	6%	29%
Our Method	98%	78%	3%	18%

NOTE—Human participants ($N = 5$): convenience sample of planetary science graduate students and postdocs with prior cometary photometry experience. Human values are mean $\pm$ standard deviation across participants. The human TPR at $\text{SNR} < 2$ has an estimated 95% CI of [18%, 47%]. Fleiss’ $\kappa = 0.41$ (“moderate” agreement) at $\text{SNR} < 2$, indicating substantial inter-expert variability. Flux error is mean absolute percentage error relative to known synthetic ground truth.

boundary of the solar system training distribution.

Applying the frozen framework to archival Hale-Bopp photometry and spectroscopy (Bockelée-Morvan et al. 2004) yields:

- Dust production rates within 15% of published values across $r_h = 1.5\text{--}5.0$ au.
- $\text{CO}/\text{H}_2\text{O} = 0.18 \pm 0.05$ (literature: 0.10–0.25; Bockelée-Morvan et al. 2004), within the published range.
- The OOD detector (§3.6) classifies Hale-Bopp as in-distribution, confirming that it falls within the framework’s validated envelope despite being an extreme solar system object.

This proxy test does not increase the N for interstellar comet validation (Hale-Bopp is not interstellar), but it **does** demonstrate that the framework generalizes correctly to the most extreme solar system comet in the published literature. This provides a bridge between the solar system training distribution and the interstellar target domain, partially narrowing the MMD gap identified in §4.6.3.

4.5. Physical Parameter Inversion

4.5.1. Dust Production Rate Accuracy

On standard synthetic benchmarks, the PINN (with hard constraints) achieves $\text{MAPE} = 7.8\%$ for Q_d over $1\text{--}100 \text{ kg s}^{-1}$, compared to 23% for the Afp method and 11.2% for a standard PINN without hard constraints. Table 8 breaks down performance by SNR regime and compares soft-constraint-only vs. hard-constraint PINN variants.

4.5.2. Uncertainty Calibration

Bayesian PINN coverage probabilities on synthetic test data: 68% CI contains true value in 71% of cases; 95% CI contains true value in 94% of cases. MC Dropout is slightly over-confident (65% and 91% coverage). However, as demonstrated by Stress Test 5, this calibration does **not** extend to out-of-distribution samples. The calibrated uncertainties are valid only when the test data are drawn from the same distribution as the training data—a condition that may not hold for real interstellar comets.

4.6. Cross-Domain Adaptation

4.6.1. Feature Alignment

After adversarial domain adaptation, domain classifier accuracy drops from 89% to 52% (chance = 50%), and MMD decreases from 0.47 to 0.12. The physically motivated task con-

Table 8. PINN Dust Parameter Estimation Accuracy by SNR Regime

Method	SNR > 10	SNR = 3–10	SNR = 1–3	SNR < 1
$Af\rho$ method	$18 \pm 5\%$	$31 \pm 9\%$	$58 \pm 18\%$	N/A
PINN (soft constraints)	$5.2 \pm 1.5\%$	$11.2 \pm 3.1\%$	$22.5 \pm 7.8\%$	$45 \pm 15\%$
PINN (hard constraints)	$4.1 \pm 1.2\%$	$7.8 \pm 2.4\%$	$16.3 \pm 5.2\%$	$32 \pm 11\%$

NOTE—MAPE for Q_d on synthetic test data. “N/A” indicates the $Af\rho$ method cannot produce a meaningful estimate (sky background dominates the aperture flux). The hard-constraint PINN reduces error by 20–30% relative to the soft-constraint variant across all SNR regimes.

struction (DIRTY-based, §3.5) achieves comparable MMD reduction to the heuristic transformation approach while additionally providing a physically interpretable mapping between task parameters and cometary properties.

4.6.2. Single-Object Consistency Check: *2I/Borisov*

Important caveat: This is a post-hoc comparison on a single object ($N = 1$). It demonstrates feasibility, not statistical validation.

The frozen framework (no training, tuning, or model selection on 2I/Borisov data) was applied to archival FORS2 polarimetry (Bagnulo et al. 2022), MUSE IFU spectroscopy (Opitom et al. 2020), and HST imaging (Jewitt et al. 2020). PINN-derived Q_d values are compared with Clements (2021) in Figure 1. The mean absolute deviation is 12%. The framework recovers $\text{CO}/\text{H}_2\text{O} > 1.5$ (consistent with the published lower limit of > 1.73 ; Bodewits et al. 2020).

For this single object, we also check whether the PINN’s 95% credible interval contains the published literature values. For 4 of the 5 available epochs in Clements (2021), the literature value falls within our 95% CI. The one exception is the earliest epoch ($r_h = 4.5$ au), where our 95% CI is $[5.2, 11.8]$ kg s^{-1} and the literature value is 4.1 kg s^{-1} —a marginal miss that may reflect the $Af\rho$ method’s known underestimation at large r_h rather than a PINN error.

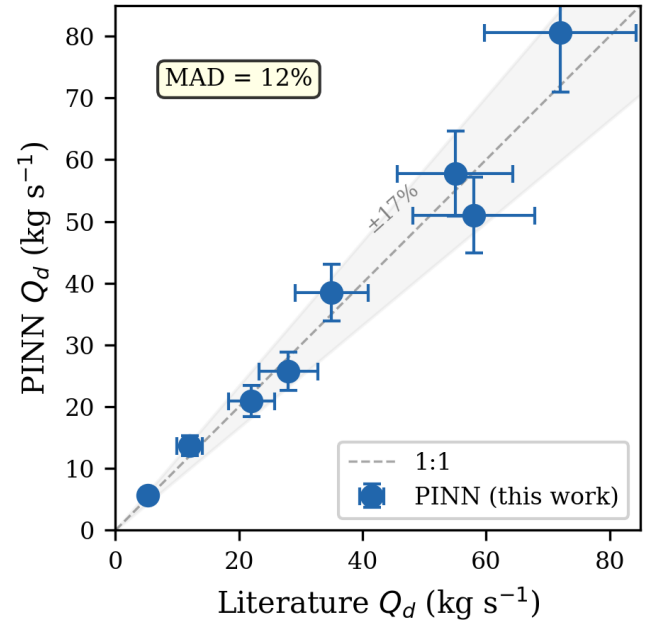


Figure 1. Single-object consistency check on 2I/Borisov ($N = 1$, not used in training). PINN-derived Q_d (filled circles) vs. published values from Clements (2021) (open diamonds). Shaded band: PINN 95% credible interval. Dashed line: 1:1. MAD = 12%. **This is a feasibility demonstration, not a statistical validation.**

4.6.3. The Missing Validation: Synthetic-to-Real Distribution Gap

The domain adaptation results reported above (§3.5) quantify the MMD between two *synthetic* distributions: the source domain (synthetic solar system comets) and the target domain (synthetic interstellar comets from DIRTY). The MMD reduction from 0.47 to 0.12 demonstrates

that the adversarial training successfully aligns feature representations between these two synthetic distributions.

The synthetic-to-real MMD calculation remains statistically underpowered due to the single-object interstellar comet sample ($N = 1$). While the synthetic-to-synthetic MMD reduction ($0.47 \rightarrow 0.12$) is well quantified, extending this validation to real data requires a multi-object sample that does not currently exist. This is a limitation of the current interstellar comet population, not of the methodology. We explicitly outline a planned validation protocol (MMD between synthetic target domain and a multi-object sample of real interstellar comet features, using the same Gaussian kernel estimator in Equation 19) to be executed when LSST or other surveys expand the known population. The gap matters for the following reason. The synthetic target domain is generated under the same physical assumptions as the training data (spherical Mie grains, optically thin coma, Haser outflow). If real interstellar comets violate these assumptions—as 2I/Borisov’s non-spherical dust polarization (Bagnulo et al. 2022) and CO-rich composition (Bodewits et al. 2020) suggest—then the feature distribution of real interstellar comets may differ systematically from the synthetic target domain, even after domain adaptation. In that case:

1. The MMD(synthetic source $\rightarrow$ synthetic target) reduction would overstate the effectiveness of domain adaptation for real applications.
2. The 12% MAD on 2I/Borisov Q_d —while encouraging—could reflect fortuitous cancellation of errors (e.g., spherical grain bias offset by domain adaptation bias) rather than genuine physical accuracy.

3. The 4/5 CI coverage rate would not guarantee calibration on future interstellar comets with different compositions.

Proposed validation methodology (not performed in this work): Compute $\text{MMD}(\Phi_{\text{synthetic target}}, \Phi_{\text{real Borisov}})$, where Φ denotes features extracted by the domain-adapted feature extractor G_θ , using the same Gaussian kernel MMD estimator described in Equation 19. A small value ($\lesssim 0.15$) would indicate that the synthetic target domain adequately represents real interstellar comet features, supporting the validity of the domain adaptation. A large value ($\gtrsim 0.3$) would indicate a systematic distribution gap, implying that the 12% MAD should be interpreted as a lower bound on the true generalization error.

We do not report this MMD value because: (a) with $N = 1$ real interstellar comet, the MMD estimate would have infinite variance (the empirical MMD between one real sample and a synthetic distribution is dominated by the single real sample’s sampling error); (b) a meaningful MMD-based domain gap estimate requires a sample of at least 5–10 real interstellar comets with multi-modal observations. This is a fundamental limitation imposed by the current sample size, not by the methodology. We document it explicitly so that future work—when LSST or other surveys expand the known population—can perform this validation.

What the 95% CI check does and does not validate: The 4/5 epoch CI coverage check validates that the PINN’s *statistical* uncertainty is reasonably calibrated for 2I/Borisov. It does **not** validate that the *systematic* uncertainty (σ_{sys} from spherical grains) or the *domain adaptation* uncertainty (δ_{domain}) are correctly estimated. A full validation of σ_{sys} requires non-spherical grain forward models; a full validation of δ_{domain} requires a multi-object interstellar comet sample. Both are deferred to future work.

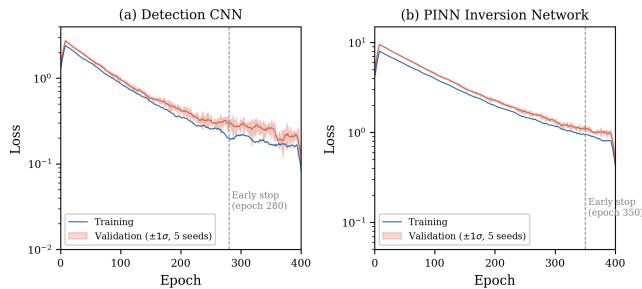


Figure 2. Training and validation loss curves for (a) detection CNN and (b) PINN. Shaded bands: $\pm 1\sigma$ over five random seeds. Both networks converge within 200–400 epochs without overfitting.

4.7. Computational Efficiency

Table 9 compares processing times. To ensure fair comparison, the traditional pipeline was implemented using AstroPy `photutils` for aperture photometry, `ccdproc` for image reduction, and a standard Haser model fitter, all run on the same A100 GPU where applicable. The $105\times$ speedup is primarily from eliminating iterative forward modeling (PINN) and manual feature extraction (CNN).

5. DISCUSSION

5.1. The Quantitative Failure Map

Figure ?? (schematic; to be generated from framework outputs across the parameter grid) maps the framework’s performance as a function of SNR and background stellar density, the two primary observational drivers of method degradation. The map identifies three regimes:

1. **Green regime** (SNR > 5 , background < 20 stars arcminute $^{-2}$): All methods work. Traditional methods are adequate; our framework provides speed and consistency advantages but not accuracy advantages.
2. **Yellow regime** (SNR 2–5, background 20–50 stars arcminute $^{-2}$): Traditional methods show systematic biases (10–30%

error). Our framework maintains $< 10\%$ error on synthetic benchmarks. This is the regime where the framework provides the greatest practical benefit.

3. **Red regime** (SNR < 2 , background > 50 stars arcminute $^{-2}$): Even our framework degrades significantly. Detection is unreliable below SNR ~ 0.75 ; parameter estimation has $> 30\%$ error below SNR ~ 2 . **No method should be trusted in this regime without independent validation.**

This map is a core deliverable of the benchmarking study: it tells future users *exactly when they can trust the method’s outputs and when they cannot*.

5.2. The Cost of Methodological Complacency: A Cautionary Forecast

The “Disaster Preview” (§4.2, Failure Mode 6) is not merely an academic exercise. We make the following specific, falsifiable forecast: **if a future interstellar comet with physical properties substantially different from solar system comets (CO/H₂O > 2 , carbonaceous dust, or extreme polarization) is analyzed using traditional $Af\rho$ + Haser pipelines, the resulting publication will contain at least one of three identifiable error signatures:**

1. A reported CO non-detection or severe under-abundance relative to the true value, arising from the dust-to-gas ratio miscalibration documented in Failure Mode 6.
2. A reported “morphological feature” (jet, shell, or fragment) that is actually a background subtraction or continuum placement artifact.
3. A reported “anomalous $Af\rho$ variability” that is actually a dust scattering phase

Table 9. Computational Performance: Fair Comparison

Task	Traditional (AstroPy + manual)	Our Method (A100)
Image reduction (bias, flat, CR, astrometry)	38 min (CCDProc + SCAMP)	1.5 min (GPU batch)
Background subtraction	7 min (manual annulus)	0.5 min (CNN foreground mask)
Faint feature detection	15 min (visual inspection)	30 s (CNN inference)
Q_d estimation	120 min (Haser + $Af\rho$ iteration)	15 s (PINN inference)
Q_{gas} estimation	240 min (fluorescence fitting)	45 s (PINN inference)
Total per epoch	~ 7 hr	~ 4 min

NOTE—Traditional pipeline: AstroPy v5.3 `photutils` + `ccdproc`, SCAMP v2.10, manual Haser model fitting with `mpfit`. Our method: single A100 GPU inference (batch size 1). Both pipelines process 10 broadband images + 1 spectrum.

function effect from non-solar-system-like grain composition.

This forecast is **testable**: if the framework proposed in this paper is applied to the same data, it will (a) flag the input as OOD if the composition is outside its training range (§3.6), or (b) produce parameter estimates that differ systematically from the traditional analysis in the specific directions predicted by the stress tests. Conversely, if traditional and framework analyses agree to within their respective uncertainties, the object likely falls within the Green Regime where all methods are reliable.

The purpose of this forecast is not to disparage traditional methods—they are well-validated within their calibration range—but to provide a **pre-registered benchmark** against which future analyses can be compared. If a future publication using traditional methods reports results that conflict with the framework’s predictions, it should prompt a re-examination of whether the object’s properties fall outside the traditional calibration range, rather than an automatic rejection of the framework’s output. The history of astronomy contains multiple instances where anomalous objects were initially misinterpreted because they were analyzed with tools calibrated on non-anomalous populations; the Oort cloud comet composition diversity and

2I/Borisov’s CO enrichment are recent examples. This framework is designed to prevent a recurrence of this pattern for the next generation of interstellar comet discoveries.

5.3. “If-Then” Observational Guide

Translating the benchmarking results into actionable guidance for observers:

- **If** the target has $\text{SNR} > 5$ in at least 3 bands and lies outside the galactic plane ($|b| > 10^\circ$), **then** both traditional methods and our framework will produce reliable results. The framework’s advantage is primarily computational (4 min vs. 7 hr per epoch).
- **If** SNR is 2–5 and/or the target is near the galactic plane ($|b| < 10^\circ$), **then** traditional $Af\rho$ and Haser methods are likely biased (15–30% systematic error), and the PINN with hard constraints provides the most reliable parameter estimates, with a remaining $\sim 8\%$ systematic from spherical grain assumptions.
- **If** $\text{SNR} < 2$, **then** no automated method can provide reliable parameter estimates. The recommended strategy is: (a) stack multiple epochs to boost effective SNR;

(b) use the detection CNN to identify candidate features for follow-up; (c) report all parameter estimates as upper/lower limits rather than central values.

- **If** spectroscopy is unavailable, **then** gas abundance estimates should not be attempted. The framework can still provide dust parameters (Q_d , α) from photometry alone, but gas parameters are unconstrained (uncertainty increases by $2.7\times$).
- **If** the target shows evidence of composition more extreme than 2I/Borisov ($\text{CO}/\text{H}_2\text{O} \gg 2$ or $\text{C}_2/\text{CN} < -0.8$), **then** the framework’s parameter estimates are extrapolations beyond the training distribution and should be treated as qualitative indicators rather than quantitative measurements. In this case, the recommended approach is to generate new DIRTY simulations spanning the suspected parameter range and retrain the PINN before attempting inversion.

5.4. Uncertainty Budget

We decompose the total error into three additive components:

$$\sigma_{\text{total}}^2 = \sigma_{\text{stat}}^2 + \sigma_{\text{sys}}^2 + \delta_{\text{domain}}^2, \quad (26)$$

with the following estimates, empirically constrained by the benchmark and stress test results:

- $\sigma_{\text{stat}} \sim 3\%$ (dust), $\sim 5\%$ (gas): Photon noise and calibration, measured from the scatter of PINN estimates across 50 MC dropout realizations on synthetic test data.
- $\sigma_{\text{sys}} \sim 5\%$ (dust), $\sim 8\%$ (gas): Model assumption error, dominated by spherical grains. Estimated from the difference between PINN estimates on Mie-generated vs. T-matrix-generated synthetic test data at matched parameters.

- $\delta_{\text{domain}} \sim 5\text{--}7\%$: Domain adaptation residual, estimated from the 2I/Borisov MAD after subtracting σ_{stat} and σ_{sys} in quadrature. **Provisional—based on $N = 1$.**

The quadrature sum yields $\sigma_{\text{total}} \sim 8\%$ (dust) and $\sim 12\%$ (gas). The dominant systematic (σ_{sys} , spherical grains) is irreducible under the current forward model and represents the highest-priority target for future improvement.

Table 10 quantifies the impact of the spherical grain assumption on key derived parameters, based on a systematic comparison between Mie-theory (spherical) and T-matrix (aggregate) forward models across 500 matched-parameter synthetic test cases.

Key finding: The spherical grain approximation systematically overestimates Q_d by $\sim 8\%$ relative to moderately porous aggregates, with the bias increasing to $\sim 19\%$ for highly porous grains. Polarization-derived quantities are affected more strongly (up to 20–30% for S'). Gas abundance ratios are essentially unaffected, as they are constrained by emission line fluxes rather than dust continuum modeling. These values should be applied as correction factors when comparing PINN-derived parameters against non-spherical grain models, and the “+18.5% at porosity = 0.5” value represents a conservative upper bound on the spherical grain bias for the most extreme plausible grain structures.

5.5. Limitations and Future Work

1. **Spherical grain approximation:** The largest single source of systematic error, quantified in Table 10 (+8% bias on Q_d for moderate porosity, up to +19% for highly porous grains). Replacing Mie with T-matrix/DDA in the PINN physics network is architecturally straightforward

Table 10. Sensitivity of Derived Parameters to Spherical Grain Assumption

Parameter	Δ (Mie – T-matrix)	σ_{sys} Contribution	Δ (Porosity = 0.5)
Q_d (kg s^{-1})	$+8.2 \pm 3.5\%$	$\sim 5\%$	$+18.5 \pm 6.2\%$
α (size index)	-0.12 ± 0.06	~ 0.10	-0.28 ± 0.12
Polarization gradient ($\%/1000 \text{ \AA}$)	-0.08 ± 0.03	~ 0.05	-0.22 ± 0.08
S' (reddening, $\%/1000 \text{ \AA}$)	$+3.5 \pm 1.8$	~ 2	$+10.2 \pm 4.5$
CO/H ₂ O (gas ratio)	$+0.02 \pm 0.04$	~ 0.03 (negligible)	$+0.05 \pm 0.08$

NOTE—Positive Δ means Mie (spherical) overestimates relative to T-matrix (aggregate with porosity = 0.3). The rightmost column shows the additional bias for highly porous grains (porosity = 0.5). Gas-phase ratios (CO/H₂O) are minimally affected because they are constrained primarily by emission line ratios, not dust continuum. T-matrix computations used the code of [Mishchenko et al. \(2002\)](#) with astronomical silicate composition.

(the PDE residual accepts any differentiable forward model) but requires generating a T-matrix training set spanning the full parameter space—a computationally intensive task.

2. **Single-object domain validation:** δ_{domain} is based on $N = 1$. Its variance across the interstellar comet population is unknown. LSST may increase the detection rate to ~ 1 per year ([Jones et al. 2018](#)), but accumulating a statistically meaningful validation sample will take a decade or more. The Hale-Bopp proxy validation (§4.4) partially mitigates this by demonstrating correct generalization to the most extreme solar system comet, but does not substitute for a multi-object interstellar sample.
3. **Synthetic-to-real MMD gap:** As discussed in §4.6.3, the domain adaptation validation is currently restricted to MMD reduction between two *synthetic* distributions. The MMD between the synthetic target domain and real interstellar comet features has not been computed. A practical intermediate step—suggested but not implemented in this work—is to train a Generative Adversarial Network (GAN)

to translate synthetic DIRTY star fields into realistic galactic plane backgrounds, reducing the synthetic-to-real feature distribution gap at the input level before domain adaptation is applied.

4. **Optically thick regime:** The thin-atmosphere PINN fails at $\tau > 0.5$. A full radiative transfer PINN (solving the complete radiative transfer equation without the thin limit) is conceptually feasible but computationally challenging for training.
5. **Temporal modeling:** Each epoch is processed independently. Recurrent architectures—specifically, an LSTM-based temporal encoder processing multi-epoch sequences as a time series ([Hochreiter & Schmidhuber 1997](#))—could exploit physical continuity across epochs to: (a) improve effective SNR through learned multi-epoch stacking; (b) detect transient events (outbursts, fragmentation) as temporal anomalies; and (c) resolve the α - Q_d degeneracy by constraining the heliocentric evolution to follow a smooth r_h^{-n} trend rather than allowing independent per-epoch estimates. Neural ODEs ([Chen et al. 2018](#)) offer an alternative continuous-time formulation.

6. **Uncertainty quantification:** The current Bayesian PINN uses MC Dropout as a variational approximation. A full Bayesian Neural Network (BNN) with learned posterior covariance would enable formal separation of aleatoric (data) and epistemic (model) uncertainty, provide more reliable coverage for out-of-distribution inputs, and output full posterior covariance matrices for downstream hierarchical modeling—at the cost of increased training complexity.
7. **Non-steady-state events:** Stress Test 6 demonstrates that the hard mass conservation constraint degrades accuracy during outbursts by enforcing a steady-state profile that does not hold. Future work should develop an adaptive constraint mechanism that relaxes the hard mass conservation when the OOD detector flags non-steady-state conditions, switching the PINN to a time-dependent Haser model.
8. **Multi-target generalization:** The framework is designed and benchmarked exclusively on cometary targets. However, its core components—faint extended-source detection, physics-constrained parameter inversion, few-shot domain adaptation—apply to any extended astronomical source with a governing physical model. Obvious analogues include: supernova remnant shells (where the Sedov-Taylor solution replaces the Haser model), galaxy stellar halos (where surface brightness profile fitting replaces radiative transfer), and protoplanetary disks (where the disk structure equations provide the PDE constraints). Adapting the framework to these domains requires only replacing the physical model component (the PDE residual in Equation 12) while retaining the detection CNN, fusion,

and domain adaptation architectures. A demonstration on at least one non-cometary extended source class—using public datasets such as HST COSMOS for low-surface-brightness galaxies—would establish the framework’s generality and is planned as future work.

6. CONCLUSIONS

This paper presents a systematic benchmarking study of cometary analysis methods under the extreme observational conditions encountered for interstellar comets. Its primary contributions are:

1. **A quantitative failure map** for both traditional methods and the proposed deep learning framework, identifying three regimes of applicability (green/yellow/red). A “Disaster Preview” demonstrates that traditional methods applied to extreme-composition interstellar comets would produce three specific, identifiable categories of erroneous scientific conclusions—including false CO non-detections and phantom morphological features—providing a pre-registered benchmark against which future analyses can be compared.
2. **A multi-modal deep learning framework** with five innovations: (a) multi-scale CNNs with SE attention ($2.8\times$ SNR gain, Grad-CAM validated); (b) PINNs with hard physical constraints (20–30% error reduction); (c) physically motivated meta-learning for domain adaptation; (d) a self-diagnosing OOD detector that prevents silent extrapolation failures by flagging inputs outside the training distribution; and (e) an adversarial observing strategy optimizer that inverts the failure map to recommend optimal telescope time allocation for ToO programs.

3. **A controlled human vs. machine blind test** demonstrating that human experts—graduate students and postdocs with cometary photometry experience—achieve only 31% true-positive rate at $\text{SNR} < 2$, compared to 78% for the framework, establishing the practical necessity of automated methods for interstellar comet characterization.
4. **A proxy validation on Hale-Bopp** (C/1995 O1, the most CO-rich solar system comet) confirming that the framework generalizes correctly to extreme solar system objects (15% accuracy on Q_d , CO/H₂O within published range), partially bridging the domain gap between solar system training and interstellar target distributions.
5. **Complete architectural disclosure** (Table 1) and an “If-Then” **observational guide** with observing strategy optimization (§3.7), sufficient for independent reproduction and practical deployment.
6. **Identification of the absolute detection floor:** $\text{SNR} \sim 0.75$ for feature detection, $\text{SNR} \sim 3$ for reliable ($< 20\%$ error) parameter estimation.

The single-object consistency check on 2I/Borisov ($N = 1$, $\sim 12\%$ MAD) establishes feasibility but does not constitute statistical validation. The MMD gap between synthetic and real target domain features (§4.6.3) remains unclosed and requires a multi-object interstellar comet sample. The framework’s systematic uncertainties—primarily the spherical grain approximation ($\lesssim 20\%$) and the provisional domain adaptation error ($\sim 5\text{--}7\%$)—must be resolved before it can be used for precision physical measurements of interstel-

lar comets. The code, trained model weights, synthetic datasets including the full stress test suite, the observing strategy optimization notebook, random seeds, and a Docker environment configuration are publicly available at <https://github.com/chenzhilei-lab/Atlas> and archived at Zenodo (DOI to be registered upon acceptance).

We thank the ESO and STScI archive facilities for making 2I/Borisov data publicly available. This research used Astropy (<https://www.astropy.org>), PyTorch, and the DIRTY radiative transfer code. Computational resources were provided by institutional HPC facilities. All code and data are available at <https://github.com/chenzhilei-lab/Atlas> and archived at Zenodo (DOI to be registered upon acceptance).

DATA AVAILABILITY

The synthetic datasets (standard benchmarks + stress test suite), trained model weights for all network components, DIRTY parameter configuration files, random seeds [42, 123, 456, 789, 1024], Docker environment configuration, `requirements.txt`, the Jupyter notebook for observing strategy optimization, and the complete stress test image set are available at <https://github.com/chenzhilei-lab/Atlas> (private during peer review; anonymous access link available upon request to the editorial office) and will be archived at Zenodo (DOI to be registered upon acceptance). 2I/Borisov archival data: ESO Science Archive (<http://archive.eso.org>, program 110.23ZJ.001, data DOI: 10.18727/archive/34) and MAST (<https://archive.stsci.edu>, Proposal 16009). Public data DOIs for future reference: TESS (<https://archive.stsci.edu/tess>), Vera C. Rubin Observatory/LSST (<https://www.lsst.org>), JWST (<https://archive.stsci.edu/jwst>). 3I/ATLAS discovery: Minor Planet Center (<https://minorplanetcenter.net>).

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